

Proximal-to-Distal Energy Transfer in the Golf Swing
Mechanisms, Evidence, and Counterfactual Tests of Distal-Handoff Timing

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Abstract

Proximal-to-distal (P→D) sequencing — the ordered peaking of pelvis, thorax, arm, and club rotational speeds — is the most widely reported organizing principle of the golf downswing, and it is usually read as evidence that mechanical energy should be passed outward along the kinetic chain as early and as completely as possible. This study reviews the mechanics and the empirical literature behind that reading and asks the sharper question: *when* during the downswing should the P→D transfer occur? Across double-pendulum mechanics, delayed-release simulation, optimal control, and the segmental kinetic-energy literature, the evidence favors a strategy that *retains* energy in the proximal system early in the downswing and *accelerates* the handoff late. We formalize this retain-early/release-late hypothesis in the language of zero-torque counterfactual (ZTCF) and zero-velocity counterfactual (ZVCF) decompositions, and test it in a reproducible two-segment golf model using an exact interaction-force decomposition, a matched-state torque-killswitch comparison, a wrist-torque timing sweep (92 torque programs), pointwise drift/control counterfactual decompositions, and Robertson-Winter style wrist-interface power accounting, all generated by committed, reproducible scripts. In this model, driving the wrist from the top produced 15% less clubhead speed than a fully passive wrist, while delaying the wrist drive to the optimal onset produced 12% more, and a program that actively restrains the wrist early and drives it late is best of all at both shoulder-torque levels tested. The energy accounting shows the winning programs move *less* energy into the club early and far more late, with the best program performing *negative* wrist work early — active retention. We interpret these results against the literature, state their limitations (a planar 2-DOF model with a fixed hub and rigid shaft). A one-at-a-time sensitivity study spanning 13 parameter cases preserves the strategy ordering but shifts the optimal onset, showing that the modeled mechanism is more robust than any single timing prescription. The force analysis resolves the geometric sine and cosine projections that determine whether distal centripetal and tangential forces transfer power, and uses the author’s prior two-hand WSCG model as a registered hypothesis source for the next level of validation. References, analysis code, machine-readable results, and provenance are available with the document for independent inspection and reuse.

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i Scope and Evidence Categories

This document distinguishes three kinds of statement. **Empirical findings** summarize published measurements of golfers. **Model-derived findings** report deterministic simulations generated by the archived analysis code and parameter sets. **Hypotheses and interpretations** state claims that remain to be tested in higher-fidelity models or human data. The distinction matters because agreement between a simplified model and prior literature is not independent empirical confirmation.

Chapter 1

Introduction

1.1 Motivation

Clubhead speed at impact is the dominant controllable determinant of drive distance, and it correlates strongly with playing standard (Fradkin, Sherman, and Finch 2004; Hellström 2009; Hume, Keogh, and Reid 2005). The golf downswing is among the most-studied examples of a *sequenced* multi-segment rotation in sport biomechanics (McPhee 2022; Dillman and Lange 1994). Across measurement systems and populations, the downswing of a skilled golfer shows a characteristic order: the pelvis reaches its peak angular speed first, then the thorax, then the lead arm, and finally the club, with each successive segment peaking faster than the one before (Cheetham et al. 2008; Tinmark et al. 2010). This pattern — the *kinematic sequence*, an instance of the general proximal-to-distal (P→D) sequencing observed in throwing and striking skills (Putnam 1993) — is routinely interpreted as the visible signature of mechanical energy generated in the large proximal segments and transferred outward to the small, fast distal ones, and it has been translated into coaching doctrine accordingly (Smith et al. 2015).

That interpretation leaves a first-order question unanswered:

Should a golfer attempt to transfer energy from proximal to distal segments *early* in the downswing, or should energy be *retained* in the proximal segments early and the transfer deliberately *accelerated* late?

The two strategies imply opposite coaching cues (“fire the club from the top” versus “hold the lag and let it release”), opposite torque programs at the wrists and torso, and different predictions for what counterfactual decompositions of a measured swing should show. The question also connects to a long-standing thread in the modeling literature: double-pendulum analyses have argued since the 1960s that the highest clubhead speeds arise when the distal joint is *not* driven early — when release is passively or actively delayed (Cochran and Stobbs 1968; T. Jorgensen 1970; Milburn 1982; Pickering and Vickers 1999) — while the coaching translation of the kinematic sequence often emphasizes initiating distal motion promptly after transition.

This study approaches the question with *counterfactual dynamics*. Two decompositions are evaluated: the **zero-torque counterfactual (ZTCF)** — the acceleration the system would exhibit at the current instant if all applied joint torques were removed, isolating what accumulated motion, gravity, and inter-segment coupling produce on their own — and the **zero-velocity counterfactual (ZVCF)** — the acceleration the applied torques, together with gravity, would produce at the current pose from rest, removing all motion-dependent effects. Preliminary torque-removal analyses motivated the hypothesis examined throughout this document: that *restraining* the P→D handoff early in the downswing and *accelerating* it late may be the more effective strategy.

1.2 Research Questions

The document is organized around four questions:

- **RQ1 — Mechanism.** By what mechanical pathways does energy generated proximally reach the club, and what does each pathway imply about the *timing* of the handoff? (Section 2)
- **RQ2 — Evidence.** Where does the P→D transfer occur in measured swings, and how do skilled and less-skilled players differ — in sequence order, in magnitudes, or in transfer efficiency?
(1)
- **RQ3 — Formalization.** How can “transfer early” versus “retain-early/release-late” be stated as falsifiable claims about measurable quantities, and what do the ZTCF/ZVCF counterfactuals predict under each? (Section 6)
- **RQ4 — Model test.** What does a two-link model show when the timing of the distal handoff is manipulated directly, with the energy accounting and counterfactual decompositions computed along every trajectory? (Section 8, Section 9, Section 10)

1.3 Approach and Contributions

The document makes six contributions:

1. a synthesis of the mechanics of sequenced swings, from the summation-of-speed principle through segment-interaction dynamics, parametric energy transfer, and optimal control (Section 2);
2. a methodological account of what “energy transfer” means operationally — joint power, joint-force power, segmental energy accounting, induced accelerations — and where the popular kinematic-sequence proxy fails (Section 2.5);
3. a critical review of the empirical evidence, including the directly relevant segmental kinetic-energy sequencing studies (Kenny et al. 2008; Anderson, Wright, and Stefanyshyn 2006) and the skill-level literature (1);
4. a formalization of the timing question and of the ZTCF/ZVCF counterfactual framework, including the distinction between pointwise and re-simulation counterfactuals (Section 6);
5. a **reproducible model test** on a double-pendulum golf model — 92 torque programs swept over wrist-drive onset time and profile, scored at a consistently defined impact, with pointwise drift/control decompositions and Robertson-Winter wrist-interface power accounting computed along every representative trajectory (Section 8, Section 9); and
6. a bridge from these 2-DOF results to broader tests: what the simplified model can and cannot establish, and the concrete, steps that extend the analysis to re-simulation parity, full-body engines, and motion-capture replication (Section 10, Section 11).

1.4 Reading Guide

Readers interested only in the new quantitative material can read Section 6 for the hypothesis, Section 8 for the setup, and Section 9 and Section 10 for the outcome. Readers using this document as a literature resource will find the review chapters (Section 2, 1) and the verified bibliography self-contained. Appendices document software and data availability (Section 12), reproducibility (Section 13), collect validation evidence (Section 14), and tabulate model parameters (Section 15).

Chapter 2

Mechanics of Sequenced Swings

This chapter assembles the mechanical foundations the rest of the document builds on: the dynamics formalism (Section 2.1), the summation-of-speed principle and its dynamical correction (Section 2.2), the double-pendulum tradition that made release timing an explicit variable (Section 2.3), the optimal-control literature (Section 2.4), and the operational meanings of “energy transfer” (Section 2.5).

2.1 Dynamics Formalism

Write the equation of motion of an articulated chain in standard manipulator form (Featherstone 2008):

$$M(q) \ddot{q} = \tau - b(q, \dot{q}), \quad b(q, \dot{q}) = C(q, \dot{q}) \dot{q} + g(q) + d(\dot{q}), \quad (2.1)$$

with configuration q , inertia matrix M , applied generalized torques τ , Coriolis/centripetal terms $C\dot{q}$, gravity g , and velocity-dependent dissipation d . Two structural facts drive everything that follows. First, the system is *coupled*: M is not diagonal, so torque at any joint induces acceleration at every joint — the basis of induced-acceleration analysis (Zajac and Gordon 1989) and of the counterfactual decompositions of Section 6. Second, the equation is *affine in τ* : the acceleration splits exactly into a drift part (what accumulated motion and gravity produce alone) and a control part ($M^{-1}\tau$), a property enforced to numerical tolerance by the accompanying tests (Section 14).

2.2 The Summation-of-Speed Principle and Its Limits

The oldest formalization of P→D sequencing is the *summation of speed* principle: in skills demanding high end-point speed, each segment should begin its motion at the moment of peak speed of the segment proximal to it, so that distal speed is built on top of proximal speed (Bunn 1972). Milburn applied the principle to golf with a double-pendulum analysis of the downswing and found the characteristic pattern of an initially dominant proximal (arm) rotation followed by rapid distal (club) rotation, noting that “*an initial delay in the uncocking of the wrist*” allows the arm to reach greater acceleration and the club’s acceleration to be summed with the proximal segment’s maximum (Milburn 1982). Note the structure of that finding: the summation principle, taken naively, prescribes *sequential initiation*; Milburn’s analysis showed the benefit comes specifically from *delaying* the distal contribution.

Putnam’s segment-interaction analyses put the principle on a dynamical footing and exposed its limits (Putnam 1991, 1993). In a linked chain, each segment’s angular acceleration depends not only on joint torques but on *motion-dependent interaction torques* — centripetal and Coriolis terms proportional to squares and products of segment angular velocities (the $C\dot{q}$ term of Equation 2.1). Putnam showed the P→D pattern in striking and throwing arises substantially from these interaction torques: proximal deceleration and distal acceleration late in the motion are coupled through the joint force and occur even without a distal joint

torque. Feltner and Dapena’s classic analysis of the baseball pitch quantified the same interplay at the shoulder and elbow (Feltner and Dapena 1986), and Herring and Chapman showed in simulated throws that the *timing* of torque onsets and torque reversals — including deliberately reversed (restraining) torques — materially changes end-point speed (Herring and Chapman 1992). Two implications matter for golf. First, kinematic order does not reveal *which* torques produced it: an identical peak sequence can arise from different mixes of muscular and interaction torques. Second, the proximal deceleration observed in experts need not be an active “braking” strategy — it can be the dynamical *consequence* of the distal segment accelerating and back-reacting on the chain.

Marshall and Elliott added a measurement caution: planar peak-ordering analyses miss long-axis rotations, which peak very late in racquet skills and contribute substantially to end-point speed (Marshall and Elliott 2000); the golf analogue is forearm rotation and wrist action in the final phase of the downswing.

2.3 Double-Pendulum Mechanics of the Golf Downswing

The two-lever, one-hinge model of the downswing — an “arm” link rotating about a hub, connected by a wrist hinge to a “club” link — has been the workhorse of golf mechanics since Cochran and Stobbs’ *Search for the Perfect Swing* (Cochran and Stobbs 1968) and Jorgensen’s dynamical analysis fitted to stroboscopic photographs of a professional swing (T. Jorgensen 1970; T. P. Jorgensen 1999); Penner’s review surveys the tradition (Penner 2003). Its enduring value here is that it makes the *timing* of the P→D handoff an explicit variable (the “release”: the point at which the wrist angle is allowed, or driven, to open) and exposes the mechanism by which delay pays. Three results anchor the analysis:

Delayed release increases clubhead speed. Pickering and Vickers, sweeping release angles in a driven double-pendulum model, found clubhead velocity at impact increases as release is delayed — the classical “late hit” (Pickering and Vickers 1999). Jorgensen’s fitted model reproduced the expert swing only with a substantially delayed opening of the wrist angle (T. Jorgensen 1970; T. P. Jorgensen 1999). In these models an early release (early P→D handoff) is strictly inferior: club kinetic energy purchased early is paid for with arm speed that never develops.

Late wrist torque helps; early wrist torque is the wrong tool. Sprigings and Neal, using a torque-driven three-segment model with physiologically bounded actuators, found that an active wrist torque applied *during the latter stages of the downswing* increased clubhead speed by approximately 4% (Sprigings and Neal 2000); Sprigings and MacKenzie confirmed the delayed-release benefit in a follow-up simulation focused on release timing (Sprigings and MacKenzie 2002). The point is not that distal effort is useless, but that its productive window is late.

Energy transfer to the club can be largely passive — governed by the changing geometry. White analyzed an *undriven* double pendulum and showed that the transfer of kinetic energy from arms to club late in the downswing is a parametric-transfer phenomenon: as the club swings out, the effective moment of inertia of the system changes, and conservation of energy and angular momentum move kinetic energy distally without any wrist torque at all; the wrist-cock angle maintained before release is the principal efficiency-determining parameter (White 2006). Miura’s analysis of the “inward pull” complements this: an expert’s hands moving toward the body near impact — shrinking the hub radius when centrifugal loading is fully developed — performs work against the centrifugal force that appears as additional clubhead kinetic energy (Miura 2001). Nesbit and McGinnis’ hub-path analyses generalize the same idea: the non-circular hand path of skilled golfers, including its late-downswing tightening, is a mechanism for shaping kinetic transfer from golfer to club (Nesbit and McGinnis 2009; Nesbit and McGinnis 2014).

Together these results describe a machine whose highest-output mode is: *load the proximal rotation first, hold the distal joint folded (retaining energy in a compact, fast-rotating configuration), then let geometry and — in the last phase — modest distal torque move the energy outward rapidly.* The double-pendulum literature is already an argument for **retain-early / release-late**. Section 9 tests exactly this proposition in the numerical implementation used here.

2.4 Optimal Control and Torque Sequencing

If the timing question is posed as an optimization — what torque program maximizes clubhead speed subject to actuator limits? — the answer has been studied repeatedly. Lampsas’s early optimal-control study of a driven double pendulum computed torque histories maximizing drive distance, and notably found the *optimal* torques differed in structure from those measured (via inverse dynamics) in his own swing (Lampsas 1975) — early evidence that human swings need not be optimal and that the optimum is a meaningful target rather than a redescription of expert behavior. Sharp, fitting shoulder-arm-club models to expert swing data and then optimizing torque programs, established a pattern for the best use of the available driving torques across the downswing phases, with the distal contribution structured late rather than distributed evenly (Sharp 2009). Three-dimensional forward-dynamics models extended the approach with anatomically meaningful actuators and flexible shafts: MacKenzie and Sprigings’ 3D model (MacKenzie and Sprigings 2009), the carry-optimized model of Balzerson, Banerjee and McPhee (Balzerson, Banerjee, and McPhee 2016), and the six-degree-of-freedom dynamic optimization of McNally and McPhee (McNally and McPhee 2018) all generate expert-like clubhead speeds with torque programs whose distal elements act late in the downswing. McPhee’s review surveys this modeling lineage (McPhee 2022).

A caution cuts the other way: optimal control optimizes *within a model*, and the sensitivity of optimal timing to model structure (hub mobility, shaft flexibility (Betzler et al. 2012; Osis and Stefanyshyn 2012), actuator rate limits) is why cross-engine validation (Section 7) is valuable when revisiting the question.

2.5 What “Energy Transfer” Means Operationally

Claims about energy transfer require an accounting framework. Four levels of analysis appear in the literature, and they are not interchangeable.

2.5.1 Joint Power and Work

For a joint spanning segments i and $i+1$ with net joint torque τ and relative angular velocity ω , the *joint (moment) power* is $P_m = \tau\omega$; its time integral over a phase is the mechanical work done by the net moment. Positive work indicates generation, negative work absorption. This is the quantity most golf studies report, and it is the basis of the work-power budget of Nesbit and Serrano (Section 5.1) (Nesbit and Serrano 2005).

2.5.2 Joint-Force Power and Segmental Energy Accounting

Joint moments are not the only conduit of mechanical energy: the joint reaction force $\mathbf{F}$ acting at a joint moving with velocity $\mathbf{v}$ transmits *joint-force power* $P_f = \mathbf{F} \cdot \mathbf{v}$ from one segment to the next. Robertson and Winter’s classic formulation balances, for each segment, the rate of change of segment mechanical energy against the sum of joint-force powers and moment powers at its proximal and distal ends, distinguishing energy *transferred between* segments from energy *generated or absorbed* at joints (Robertson and Winter 1980; Winter 2009). This is the correct level of analysis for the timing question, because “retain-early / release-late” is precisely a claim about the *time course of inter-segment transfer terms*, not about peak angular velocities. Section 8 implements exactly this accounting at the model’s wrist interface, including the energy-balance residual check.

2.5.3 Induced Accelerations and Dynamic Coupling

In a coupled chain, any single torque induces accelerations at *every* joint — including joints it does not span — through the inverse inertia matrix (Zajac and Gordon 1989). Induced-acceleration and power-transfer analyses in locomotion research (Zajac, Neptune, and Kautz 2002) show that intuitions of the form “muscle X accelerates joint Y” routinely fail in coupled systems. For golf this matters because a torso torque early in the downswing partly accelerates the *club* (through coupling), and a wrist torque late in the downswing back-reacts on the torso; the net energy routing is an emergent property of the whole chain. The ZTCF/ZVCF decompositions of Section 6 are the counterfactual cousins of this analysis.

2.5.4 Kinematic Sequence as a Proxy — And Its Failure Modes

Peak-ordering of segment angular speeds is a *proxy* for energy flow, not a measurement of it. Marsan and colleagues showed that the identified kinematic sequence in golf depends materially on which angular-velocity component (or norm, or projection) is analyzed: thirteen golfers analyzed with seven component-selection methods yielded different sequences (Marsan et al. 2019). Marshall and Elliott’s long-axis-rotation caution (Marshall and Elliott 2000) is a special case of the same problem, and the near-planarity of the downswing is itself only an approximation (Coleman and Rankin 2005; Kwon et al. 2012). Any serious test of the timing hypothesis must therefore run at the level of energy and power accounting (Section 2.5.2), with the kinematic sequence reported as corroborating description only — which is how Section 9 uses it.

Chapter 3

Interaction-Force Mechanisms in the Double Pendulum

The phrase *energy transfer* is often used as if energy were a substance that waits in one link and is then released into the next. The mechanics are more specific. A distal segment gains mechanical energy when forces acting at its moving joint and moments acting on it do positive power. Those forces can grow without a large distal actuator torque because the segment must satisfy the acceleration constraints imposed by the moving chain. Their capacity to do work depends on orientation: a large force perpendicular to joint velocity transmits no instantaneous power, while a smaller force aligned with that velocity can be effective. This chapter derives those statements exactly for the double pendulum, tests them numerically, and identifies which conclusions survive when the commanded torques are removed.

3.1 The Question in Operational Form

Consider the club segment alone. It is acted on by gravity, a force at the wrist, and a net wrist moment. Its kinetic-energy rate is

$$\dot{T}_2 = \underbrace{\mathbf{F}_W \cdot \mathbf{v}_W}_{\text{joint-force power}} + \underbrace{M_W \omega_2^{\text{abs}}}_{\text{joint-moment power}} + \underbrace{m_2 \mathbf{g} \cdot \mathbf{v}_{C2}}_{\text{gravity power}}, \quad (3.1)$$

where $\mathbf{F}_W$ is the wrist force acting on the club, $\mathbf{v}_W$ is the velocity of the wrist point, M_W is the net moment acting on the club at the wrist, and $\omega_2^{\text{abs}} = \dot{\theta}_1 + \dot{\theta}_2$ is the club's absolute angular velocity. Equation 3.1 separates three questions:

1. What creates a large wrist force?
2. What orients that force so it has a useful projection onto wrist velocity?
3. What part of the resulting club-energy gain requires commanded wrist torque?

The first two questions concern dynamic coupling and geometry. The third is a counterfactual question. None can be answered from angular-speed peak order alone (Putnam 1991; Robertson and Winter 1980; Winter 2009).

3.2 Coordinate and Sign Conventions

The arm angle θ_1 and relative wrist angle θ_2 are measured from the downward vertical and from the arm, respectively. The club absolute angle is $\phi = \theta_1 + \theta_2$. In swing-plane Cartesian coordinates, x points toward the target side and y points upward. Define

$$\mathbf{u}(\theta) = \begin{bmatrix} \sin \theta \\ -\cos \theta \end{bmatrix}, \quad \mathbf{u}_\perp(\theta) = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}. \quad (3.2)$$

$\mathbf{u}$ points along a link and $\mathbf{u}_\perp$ points in its positive angular-velocity direction. Figure 3.1 declares the free-body convention used in every calculation. $\mathbf{F}_W$ is the force *on the club*; the equal-and-opposite force acts on the arm. Positive $\mathbf{F}_W \cdot \mathbf{v}_W$ therefore means energy crosses the wrist interface into the club.

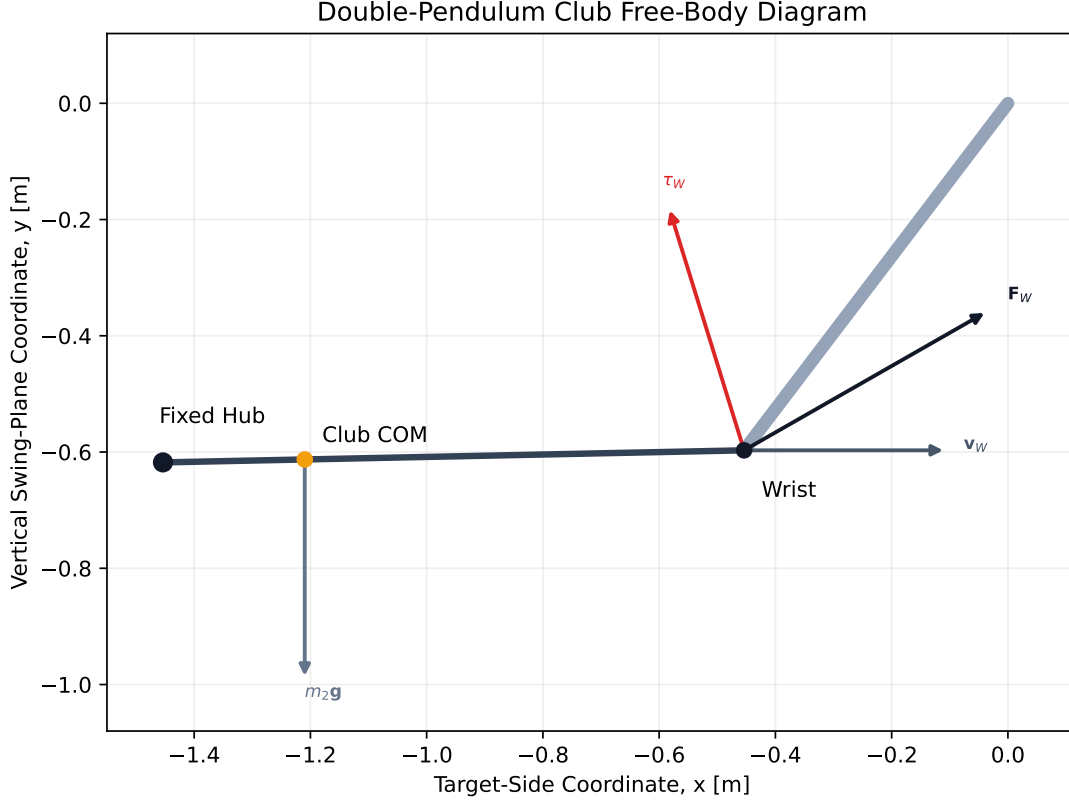


Figure 3.1: Double-Pendulum Club Free-Body Diagram

The force and moment are related but not interchangeable. Translation of the club centre of mass depends on the net force. Rotation about the centre of mass depends on both the wrist-force moment arm and M_W :

$$I_{2,C}\ddot{\phi} = (\mathbf{r}_W - \mathbf{r}_C) \times \mathbf{F}_W + M_W. \quad (3.3)$$

The executable validation checks Equation 3.3 at machine precision. This distinction becomes central in a two-hand model, where two hand forces can have a modest resultant force but a large equivalent couple.

3.3 Exact Decomposition of the Wrist Reaction Force

The wrist and club centre-of-mass positions are

$$\mathbf{r}_W = l_1 \mathbf{u}(\theta_1), \quad \mathbf{r}_{C2} = \mathbf{r}_W + l_{c2} \mathbf{u}(\phi). \quad (3.4)$$

Differentiating twice gives four acceleration terms:

$$\mathbf{a}_{C2} = \underbrace{l_1 \ddot{\theta}_1 \mathbf{u}_\perp(\theta_1)}_{\text{proximal tangential}} - \underbrace{l_1 \dot{\theta}_1^2 \mathbf{u}(\theta_1)}_{\text{proximal centripetal}} + \underbrace{l_{c2} \ddot{\phi} \mathbf{u}_\perp(\phi)}_{\text{distal tangential}} - \underbrace{l_{c2} \dot{\phi}^2 \mathbf{u}(\phi)}_{\text{distal centripetal}}. \quad (3.5)$$

Newton's second law for the club yields

$$\mathbf{F}_W = m_2(\mathbf{a}_{C2} - \mathbf{g}). \quad (3.6)$$

Combining Equation 3.5 and Equation 3.6 produces five exact force components. They are not fitted regressors and do not depend on numerical differentiation.

Table 3.1: Exact Wrist-Force Components in the Planar Double Pendulum

Component	Vector Expression	Physical Meaning
Proximal tangential	$m_2 l_1 \ddot{\theta}_1 \mathbf{u}_\perp(\theta_1)$	Force needed because the wrist has tangential acceleration
Proximal centripetal	$-m_2 l_1 \dot{\theta}_1^2 \mathbf{u}(\theta_1)$	Force needed because the wrist follows a curved path
Distal tangential	$m_2 l_{c2} \ddot{\phi} \mathbf{u}_\perp(\phi)$	Force associated with club angular acceleration about the wrist
Distal centripetal	$-m_2 l_{c2} \dot{\phi}^2 \mathbf{u}(\phi)$	Force associated with the club COM curving about the wrist
Gravity reaction	$-m_2 \mathbf{g}$	Wrist support needed in addition to gravity

Two cautions prevent common misreadings. First, *centripetal* describes an acceleration direction, not a separate applied agent. The joint reaction force is the physical constraint force; the named terms explain why that force is required. Second, gravity appears twice in a causal discussion: it contributes to the system accelerations through the equations of motion, and the explicit $-m_2 \mathbf{g}$ term is required when the club free-body equation is solved for the wrist force. These roles must not be double-counted.

3.4 Why Zero Commanded Torque Does Not Imply Zero Force

For generalized coordinates $q = [\theta_1, \theta_2]^T$, the dynamics are

$$\ddot{q} = M(q)^{-1} \{ \tau - b(q, \dot{q}) \}. \quad (3.7)$$

Setting $\tau = 0$ leaves $\ddot{q}_0 = -M^{-1}b$, which is generally nonzero whenever gravity, velocity-dependent coupling, or damping is present. Substitution of $\ddot{q}_0$ into Equation 3.5 yields a nonzero centre-of-mass acceleration and therefore a nonzero wrist reaction force. This is the precise mechanical basis for the WSCG presentation's statement that zero joint torque does not mean zero interaction force (Olson 2024). It also follows directly from standard linked-segment dynamics (Putnam 1991, 1993; Featherstone 2008).

Because the equations are affine in torque, the *instantaneous* acceleration has an exact split:

$$\ddot{q}_{\text{total}} = \underbrace{-M^{-1}b}_{\ddot{q}_{\text{drift}}} + \underbrace{M^{-1}\tau}_{\ddot{q}_{\text{control}}}. \quad (3.8)$$

At a fixed state, all velocity-squared terms in Equation 3.5 are shared between the two cases. The force split is therefore

$$\mathbf{F}_{\text{total}} = \mathbf{F}_{\text{drift}} + \mathbf{F}_{\text{control}}, \quad (3.9)$$

where $\mathbf{F}_{\text{drift}}$ is evaluated at zero commanded torque and $\mathbf{F}_{\text{control}}$ is the change induced by the commanded torque. This is a local superposition at one state. Once two futures are integrated, their states diverge and the equality cannot be applied across time by subtracting unrelated samples.

Figure Figure 3.2 shows this decomposition at six phases of the reference swing. The force scale is fixed across panels. The pointwise zero-torque force becomes large late even though the commanded torque is set to zero in that calculation; accumulated angular velocity and changing geometry continue to demand a substantial wrist constraint force.

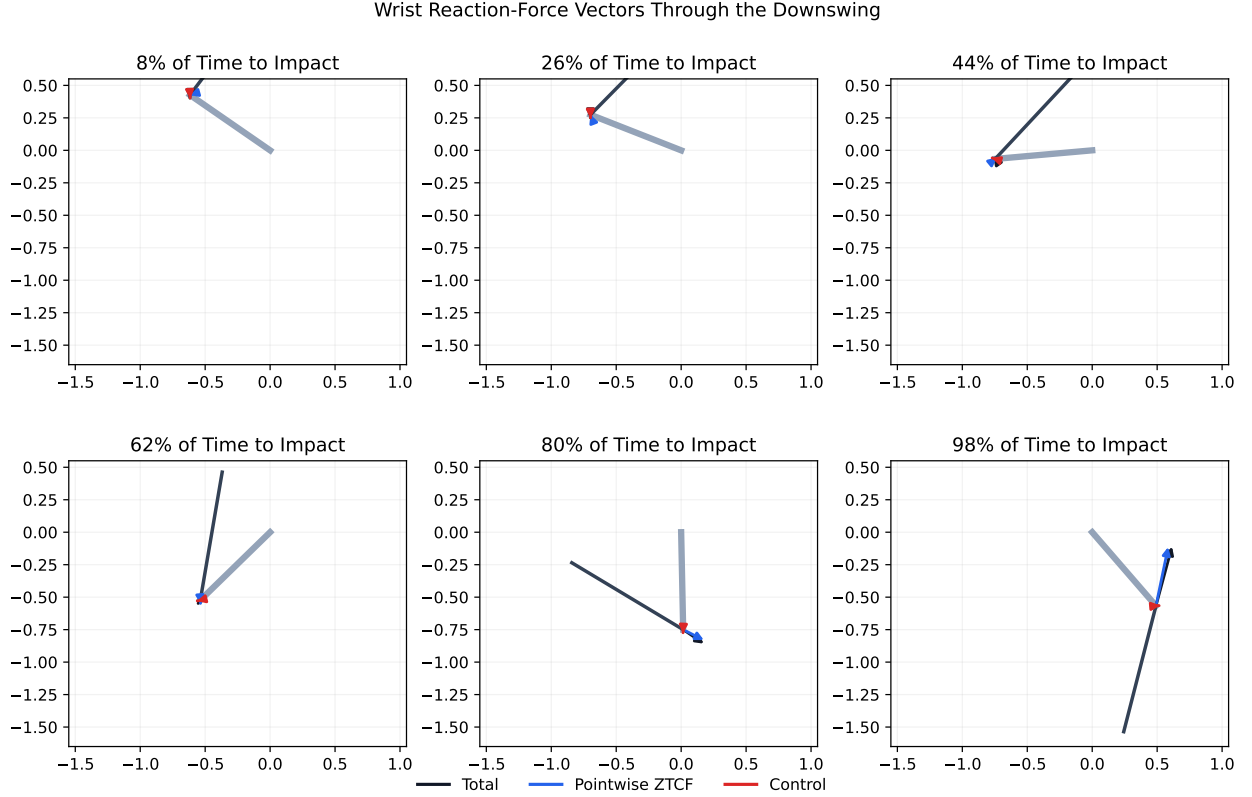


Figure 3.2: Wrist Reaction-Force Vectors Through the Downswing

3.5 Force Magnitude Is Not Transfer

The wrist velocity is

$$\mathbf{v}_W = l_1 \dot{\theta}_1 \mathbf{u}_\perp(\theta_1). \quad (3.10)$$

For any component $\mathbf{F}_k$, its contribution to club energy is $P_k = \mathbf{F}_k \cdot \mathbf{v}_W$. The dot product, not force magnitude, decides transfer. Substitution of the five components produces

$$\begin{aligned}
P_{\text{prox,tan}} &= m_2 l_1^2 \ddot{\theta}_1 \dot{\theta}_1, \\
P_{\text{prox,cen}} &= 0, \\
P_{\text{dist,tan}} &= m_2 l_1 l_{c2} \ddot{\phi} \dot{\theta}_1 \cos \theta_2, \\
P_{\text{dist,cen}} &= -m_2 l_1 l_{c2} \dot{\phi}^2 \dot{\theta}_1 \sin \theta_2, \\
P_{g\text{-reaction}} &= m_2 g_{\text{proj}} l_1 \dot{\theta}_1 \sin \theta_1.
\end{aligned} \tag{3.11}$$

Equation Equation 3.11 gives several strong results.

The proximal centripetal term does no work at a fixed circular hub path. It is radial to the arm while wrist velocity is tangential. It can be large in a force plot and still contribute exactly zero joint-force power. A mobile or noncircular hub breaks that orthogonality, which is why hand-path shaping is a separate mechanism in more complete models (Miura 2001; Nesbit and McGinnis 2009; Nesbit and McGinnis 2014).

The distal centripetal term is geometry gated. Its projection coefficient is $-\sin \theta_2$. With the club approximately 90° behind the arm ($\theta_2 \approx -90^\circ$), the coefficient is near $+1$: a force directed toward the club's instantaneous centre can project strongly along hand motion. As the links align, this coefficient falls to zero.

The distal tangential term has the complementary gate. Its coefficient is $\cos \theta_2$: near a 90° wrist cock its projection is small, while it becomes large as the links align. The changing wrist angle therefore rotates the dominant transfer pathway from one inertial component toward another.

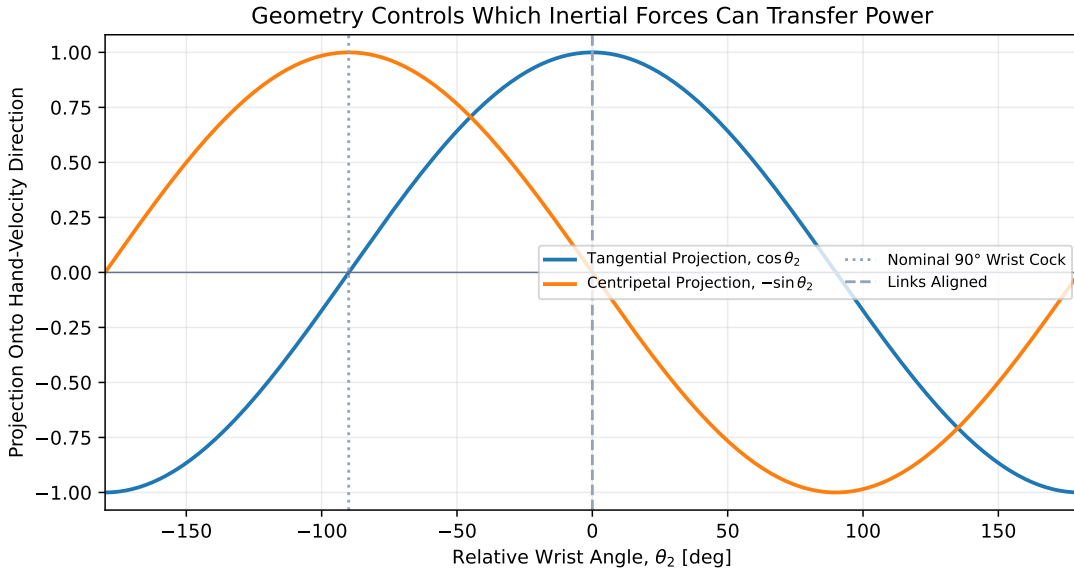


Figure 3.3: Geometry Controls Which Inertial Forces Can Transfer Power

This complementarity is the geometric core of the handoff. Early in the downswing, a compact configuration permits high proximal angular acceleration without forcing the club to acquire its final angular velocity immediately. As the wrist opens, the projection factors change while $\dot{\phi}^2$ grows. The force can therefore rise rapidly and rotate into a direction that performs positive work at the wrist. White's parametric-transfer description and Miura's hand-path analysis are consistent with this power-level account, but they emphasize different model freedoms (White 2006; Miura 2001).

3.6 Numerical Mechanism Audit

The numerical audit uses the existing reference program: $60 \text{ N} \cdot \text{m}$ at the shoulder, $-10 \text{ N} \cdot \text{m}$ at the wrist until 0.10 s , and $+15 \text{ N} \cdot \text{m}$ thereafter. Impact is the first valid upward crossing of club vertical. The program

reaches that event at 0.3493 s with 38.85 m/s clubhead speed. These values describe the declared model, not a human normative target.

The exact force reconstruction reaches a peak wrist-force magnitude of 315.4 N before impact. The pointwise zero-torque case reaches 310.6 N, while the root-mean-square magnitude of the control-force increment is 16.3% of the root-mean-square total-force magnitude. The similarity is not evidence that torque is unimportant: prior torque shaped the state, and the present torque still changes the future. It does show that the late force demand at the observed state is dominated by the moving system’s inertial and gravitational drift rather than by the instantaneous control increment.

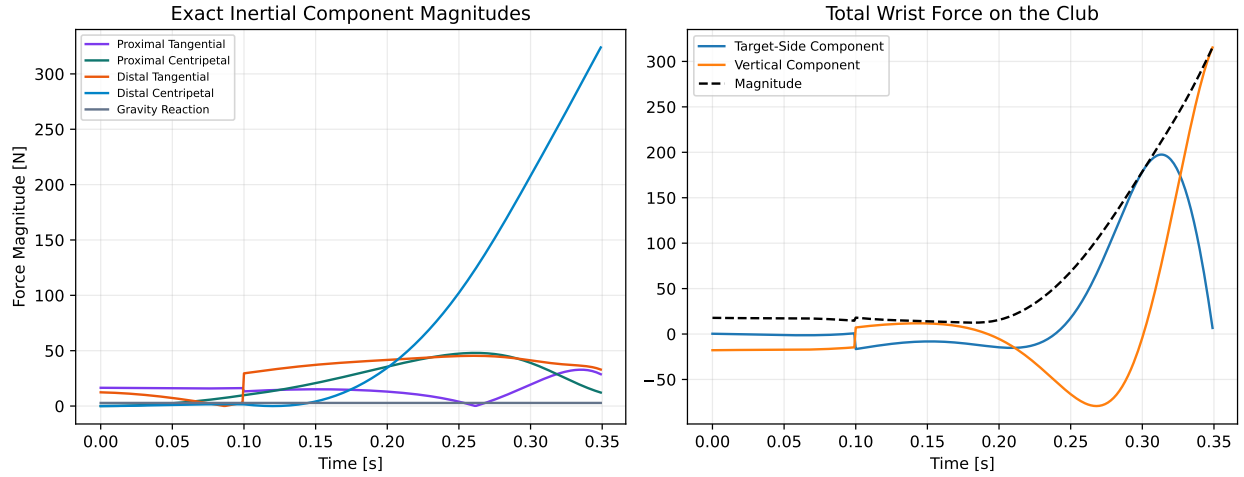


Figure 3.4: Exact Inertial Components and Total Wrist Force

Figure Figure 3.4 exposes the mechanism behind the late rise. The distal centripetal term grows approximately with $\dot{\phi}^2$ and is the largest component near impact. Its effect on transfer is then determined by the wrist-angle projection in Equation 3.11. The discontinuity in tangential terms at 0.10 s is expected: the prescribed wrist torque switches from restraint to drive, changing angular acceleration instantaneously while position and velocity remain continuous.

The integrated wrist-force work to impact is 130.7 J. The late half contributes 131.7 J, because the early half contains small positive and negative exchanges that nearly cancel. Across the full interval, positive force work is 136.8 J and negative force work is -6.1 J. The pointwise zero-torque force integrates to 132.2 J along the *commanded states*. That last quantity is an attribution diagnostic, not the energy of a realizable zero-torque future.

The force-power traces clarify why peak-force comparisons alone are insufficient. The proximal centripetal magnitude is visible in Figure 3.4 but its power trace remains zero by geometric orthogonality. The distal centripetal pathway grows into the dominant positive term late. The distal tangential pathway is negative through much of the middle downswing and turns positive near impact. Total transfer is their signed sum with proximal tangential and gravity-reaction contributions.

3.7 Pointwise and Forward Counterfactuals

Two zero-torque questions are useful, but they must have different names.

Pointwise ZTCF asks: *At this observed state, what acceleration and reaction force would the equations require if commanded torques were zero right now?* It holds $(q, \dot{q})$ fixed independently at every recorded sample. This is an instantaneous attribution and gives the exact affine split in Equation 3.9.

Matched-state torque killswitch asks: *Starting from this state, what future trajectory follows if torque remains zero?* It initializes commanded and zero-torque rollouts at the same $(q, \dot{q}, t)$ and integrates both.

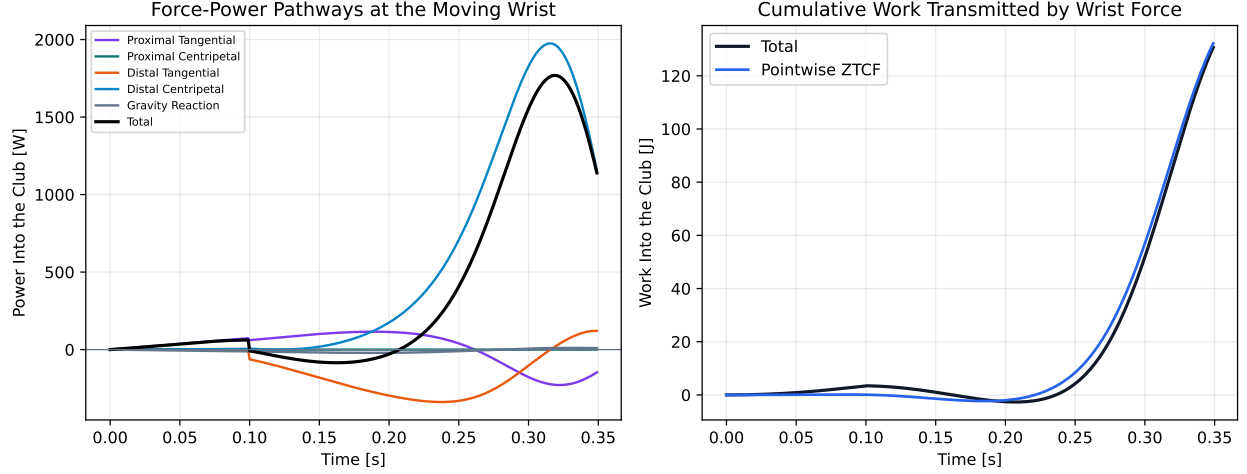


Figure 3.5: Force-Power Pathways and Cumulative Wrist-Force Work

Only the initial state is matched. As soon as accelerations differ, velocity, geometry, force, and power histories diverge.

The WSCG presentation constructs a trajectory-level counterfactual by applying kill switches at successive states and compiling the resulting values (Olson 2024). The present implementation makes the distinction executable. At a 0.251 s cut, the two rollouts begin with identical position and velocity, and the initial zero-torque acceleration exactly matches the pointwise drift calculation. After 0.120 s, their generalized positions differ by 0.540 rad and their generalized velocities by 8.49 rad/s. Figure 3.6 shows why a pointwise ZTCF trace cannot be interpreted as a torque-free replay of the measured motion.

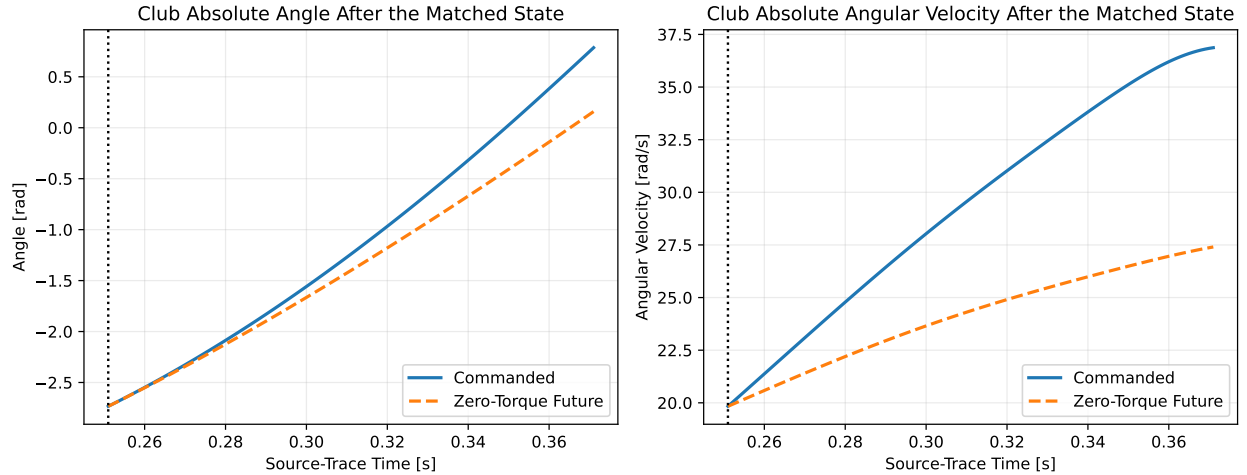


Figure 3.6: Commanded and Zero-Torque Futures From a Matched Late-Downswing State

Both counterfactuals are informative. The pointwise version isolates the force required by the state already achieved. The killswitch version tests whether that state can carry the club forward without continued control. A robust claim of passive late-swing transfer should be supported by both: large favorable pointwise drift power and persistence of the predicted behavior over a stated killswitch horizon.

3.8 Implications for Clubhead-Speed Strategy

Within this model class, the mechanism suggests a conditional strategy rather than a universal timing command.

1. **Build proximal motion while the system is compact.** Early shoulder work increases $\dot{\theta}_1$ and establishes the state on which later inertial forces depend. Premature distal drive can open the wrist before the favorable velocity-squared terms have grown.
2. **Preserve a useful relative angle long enough for inertial loading to develop.** Near a 90° lag angle, the distal centripetal force has a strong projection onto hand velocity. This is a mechanical condition, not a claim that a fixed angle should be copied across golfers.
3. **Allow geometry to redirect the transfer pathway.** As the wrist opens, centripetal projection falls and tangential projection rises. A productive release coordinates this rotation with increasing club angular speed.
4. **Use late wrist moment as a supplement, not as the sole explanation.** The tested late drive improves speed, yet most wrist-force work at the achieved states remains in the zero-torque drift term. Active torque helps create and steer the state; it need not directly account for most late transfer power.
5. **Judge a strategy at impact under constraints.** Maximizing an intermediate force, angular acceleration, or lag angle can delay delivery or create an invalid second-pass solution. Clubhead speed, orientation, actuator bounds, and robustness must be evaluated together ([Pickering and Vickers 1999](#); [Springs and Neal 2000](#)).

These conclusions are model-conditional. A mobile hub adds radial hand velocity, a flexible shaft adds stored elastic energy, three-dimensional motion adds long-axis rotations, and a biological controller adds torque-rate and activation constraints. Each extension can change the best timing while leaving the power identity $P = \mathbf{F} \cdot \mathbf{v}$ intact.

3.9 Bridge to Two-Hand Negative Coupling

A single wrist point cannot produce an independent force couple: force and wrist moment are separate inputs. Two hands change the wrench geometry. If the lead and trail hands apply $\mathbf{F}_L$ and $\mathbf{F}_T$ at positions $\mathbf{r}_L$ and $\mathbf{r}_T$, their equivalent wrench about a reference point O is

$$\mathbf{F}_{\text{net}} = \mathbf{F}_L + \mathbf{F}_T, \quad M_O = (\mathbf{r}_L - \mathbf{r}_O) \times \mathbf{F}_L + (\mathbf{r}_T - \mathbf{r}_O) \times \mathbf{F}_T. \quad (3.12)$$

Opposing hand-force components can cancel in $\mathbf{F}_{\text{net}}$ while adding in M_O . Consequently, a negative equivalent couple near impact does not by itself prove an actively commanded negative wrist torque. It can arise as the pair of constraint forces needed to enforce two moving hand contacts on the club. The WSCG two-hand planar simulation reported precisely this pattern: similar BASE and zero-torque net club forces, substantial opposing local hand forces, and a late reversal of the equivalent couple ([Olson 2024](#)).

The registered source series are redrawn in Figure 3.7. They preserve the reported BASE and counterfactual hand-force histories; they are not claimed as a reproduction of the underlying Simscape simulation.

This evidence motivates the next model test. The two-hand study must compute the equivalent wrench from local forces, verify sign conventions through power and moment balance, compare pointwise and killswitch counterfactuals, and separate momentum, gravity, shaft flex, damping, and numerical effects. The passive-negative-couple hypothesis survives only if the sign reversal persists under those controls and across reasonable contact spacing, hub paths, and shaft parameters.

3.10 Claims, Nonclaims, and Falsification Tests

The double-pendulum evidence supports four narrow claims:

- substantial wrist reaction force can exist at zero commanded torque;

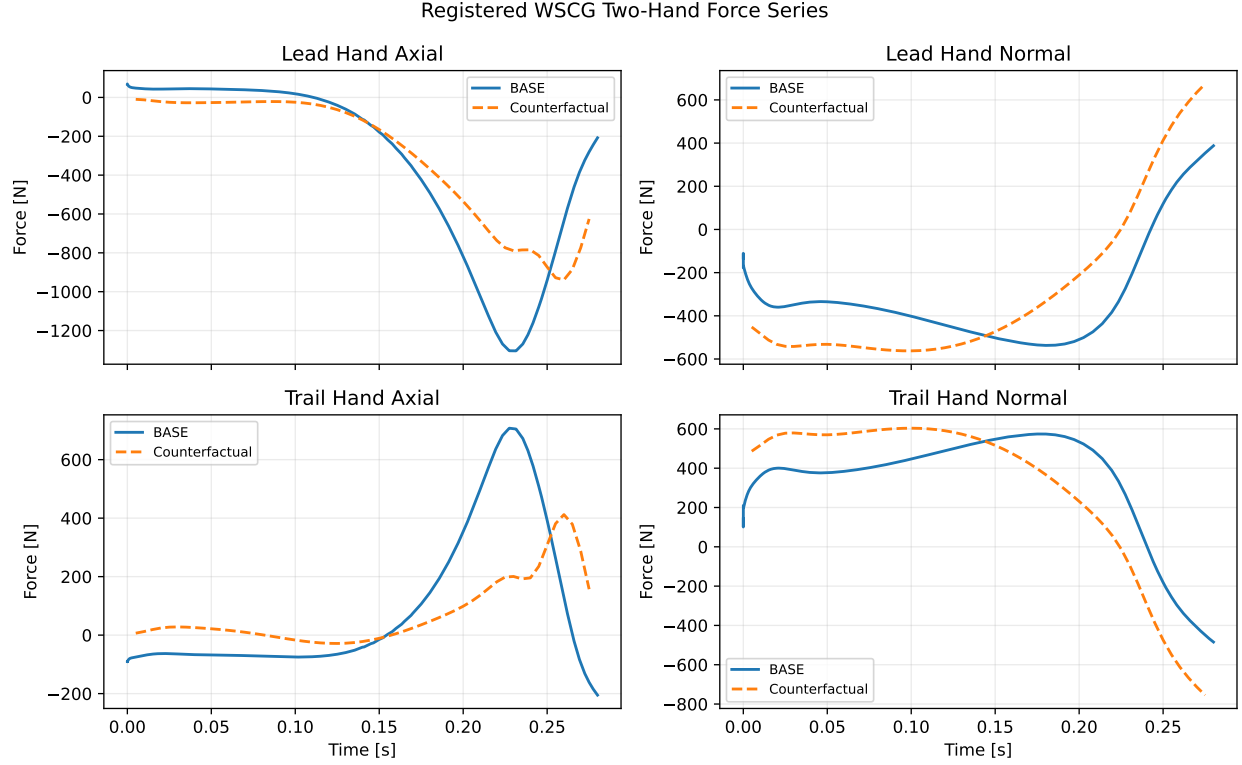


Figure 3.7: Registered WSCG Two-Hand BASE and Counterfactual Hand-Force Series

- force transfer is governed by a signed velocity projection, not force magnitude;
- wrist geometry produces exact sine and cosine gates on two major inertial power terms; and
- in the reference program, most positive wrist-force work occurs late and the pointwise drift force dominates the total-force magnitude at achieved states.

It does **not** establish that human golfers use the same torque history, that the two-hand negative couple is always passive, that more lag is always better, or that the reported parameter values are population estimates. The following tests could overturn or materially qualify the framework:

- a matched-state killswitch ensemble showing that favorable pointwise drift does not persist long enough to affect delivery;
- a two-hand wrench audit in which the negative couple disappears after gravity, shaft elasticity, damping, and numerical artifacts are separated;
- higher-order models in which hub translation or three-dimensional rotation reverses the dominant power pathway; or
- synchronized human motion and force estimates inconsistent with the modeled timing and sign structure.

Stating those failure conditions is part of the scientific result. The present double-pendulum treatment identifies a mechanism and supplies reproducible tests; it does not promote a software package or convert model behavior into a coaching prescription.

Chapter 4

Matched-State Counterfactual Persistence

The previous chapter established an instantaneous result: at a fixed state, zero commanded torque can coexist with substantial wrist reaction force and positive joint-force power. That result does not show how long the behavior persists. This chapter tests persistence by releasing the commanded and zero-torque systems from identical states at multiple downswing phases and integrating both futures over controlled horizons.

4.1 Experimental Contract

Let the source trace provide a state $x_c = [q(t_c), \dot{q}(t_c)]$ at cut time t_c . Two simulations are initialized with exactly that state:

$$x_{\text{cmd}}(0) = x_0(t_c) = x_c. \quad (4.1)$$

The commanded future continues the declared shoulder and wrist torque program; the killswitch future uses $\tau = 0$ at both joints. The initial acceleration difference is

$$\ddot{q}_{\text{cmd}}(0) - \ddot{q}_0(0) = M(q_c)^{-1} \tau(t_c). \quad (4.2)$$

At the initial instant, the zero-torque acceleration, force, and force power are the pointwise ZTCF quantities. At every later instant, each future is evaluated on its own state. This construction preserves causal meaning:

- pointwise ZTCF attributes acceleration and force at the state already reached;
- the killswitch tests whether the state carries the passive mechanism forward; and
- commanded-minus-killswitch differences quantify the consequences of continued control over a declared horizon.

The ensemble spans eight cut times from 0.080 to 0.320 s, including 0.099, 0.100, and 0.101 s around the wrist-torque switch; four horizons from 20 to 120 ms; and integration steps of 0.5, 1, and 2 ms. This gives 96 baseline comparisons. Every row records position, velocity, wrist force, wrist-force power, integrated force work, and terminal clubhead-speed differences.

4.2 Why Cut Time and Horizon Matter

A counterfactual effect is not a property of a trajectory in the abstract. It depends on *when* the intervention occurs and *how long* the futures are allowed to diverge. A short early killswitch mostly tests the immediate

acceleration change. A long late killswitch also changes geometry, club angular velocity, and the velocity-squared force terms. Figure 4.1 shows this dependence directly.

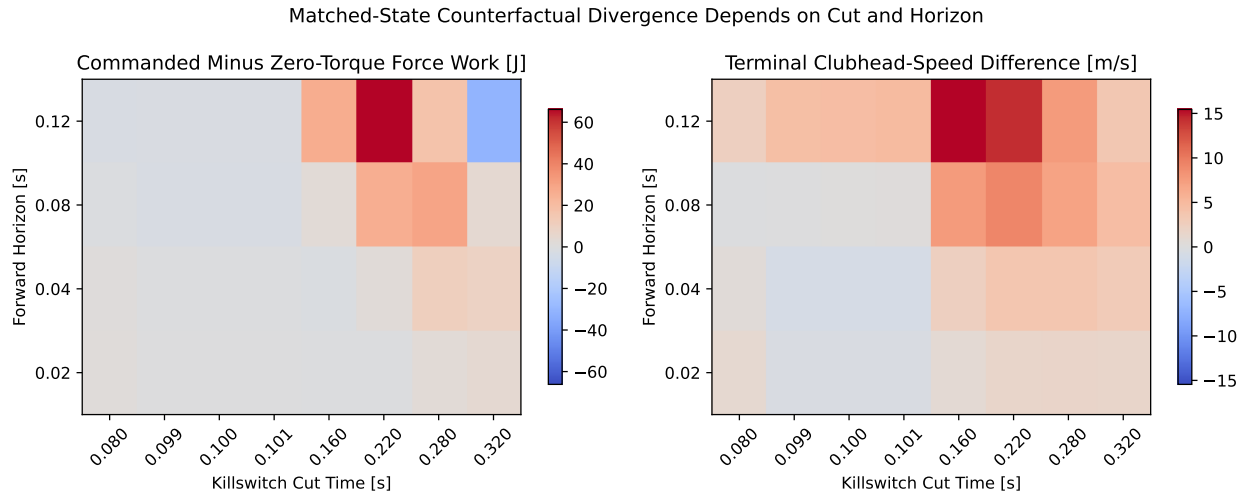


Figure 4.1: Matched-State Counterfactual Divergence by Cut Time and Horizon

At the 1 ms reference step, commanded-minus-zero-torque wrist-force work ranges from -30.88 J for a 0.320 s cut followed for 120 ms to +66.17 J for a 0.220 s cut over the same horizon. The corresponding terminal clubhead-speed differences are +3.42 and +14.20 m/s. A negative force-work difference does not mean the commanded future is globally slower: the wrist moment, gravity, and the state inherited at the cut also contribute to club energy. It means only that, over that horizon, continued control produces less *wrist-force work* than the zero-torque future.

The maps therefore reject a binary interpretation such as “the late swing is passive” or “continued torque is necessary.” Early and middle cuts generally show that continued control creates a rapidly different state and often more clubhead speed. Very late cuts can show less commanded force work because the two futures encounter different orientation and deceleration histories. The mechanism is phase dependent.

4.3 Separation of State, Force, and Power

Figure 4.2 follows three representative cuts for 80 ms. The curves coincide at the vertical dotted line by construction. Their slopes then separate because acceleration differs. Force and power can diverge faster than angle because both depend directly on acceleration and velocity-squared terms.

The 0.12 s cut occurs soon after the restraint-to-drive transition. Continued wrist drive produces a more open club state, but commanded wrist-force power is initially more negative than in the zero-torque future. At 0.22 s, commanded force and power rise much faster than the killswitch values, producing a large positive work difference. At 0.30 s, both futures initially retain high positive power because both inherit a rapidly moving club. This last panel is the clearest persistence example: the zero-torque future does not lose its large interaction force when torque is removed; it loses ground gradually as the state evolves differently.

This behavior clarifies the meaning of passive contribution. A passive term can be large because earlier active work created high velocity. Removing torque late does not erase that history. Conversely, observing a large zero-torque force at the cut does not show that the same force would persist indefinitely. The matched-state horizon is part of every claim.

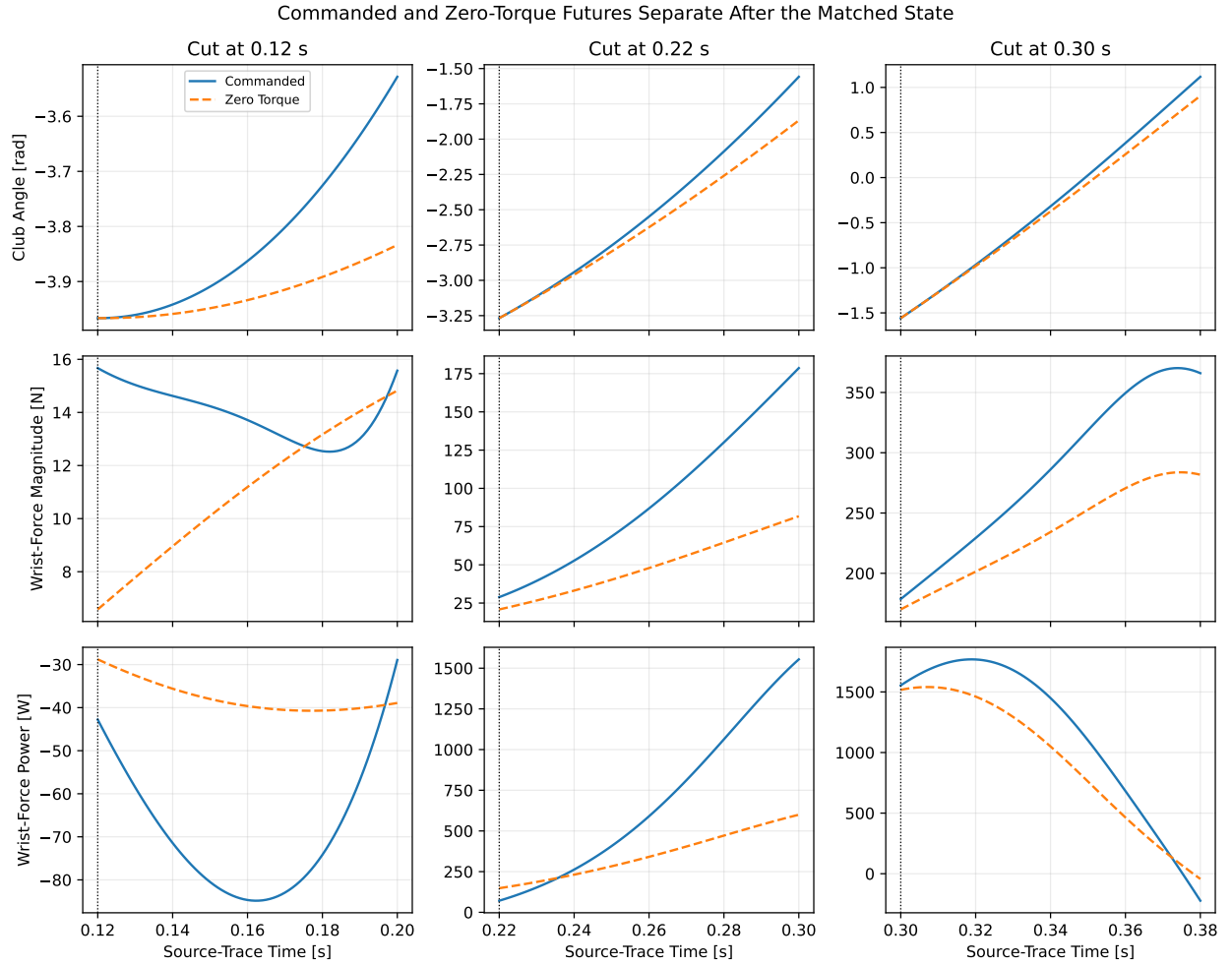


Figure 4.2: Commanded and Zero-Torque State, Force, and Power Futures

4.4 Control Discontinuity Audit

The prescribed wrist torque changes from -10 to $+15 \text{ N} \cdot \text{m}$ at 0.100 s . A pointwise split changes discontinuously at that time because the control vector changes, while position and velocity remain continuous. Three cut times bracket the switch. Over an 80 ms future, commanded-minus-zero-torque force work is -2.83 J at 0.099 s , -2.87 J at 0.100 s , and -2.87 J at 0.101 s ; terminal clubhead-speed differences are -0.03 , $+0.01$, and $+0.14 \text{ m/s}$. The force-distance metric changes smoothly from 19.98 to 20.85 N .

The close neighboring results show that the ensemble is not dominated by a single-sample numerical impulse. The acceleration changes discontinuously as the idealized step command requires, but RK4 integrates finite state changes, and the adjacent-cut futures remain continuous at the reported scale. A physiological torque-rate model remains preferable for human interpretation; the present audit establishes numerical behavior for the declared open-loop program only.

4.5 Timestep Convergence

The 0.5 ms result is used as the reference. Across all cut-time and horizon combinations, the maximum 1 ms difference is $3.9 \times 10^{-10} \text{ rad}$ in terminal angle distance, $1.1 \times 10^{-8} \text{ rad/s}$ in terminal velocity distance, $1.1 \times 10^{-7} \text{ N}$ in terminal force distance, 0.0174 J in force-work difference, and $4.5 \times 10^{-9} \text{ m/s}$ in terminal clubhead-speed difference. The 2 ms step produces larger but still bounded differences: up to 0.0091 rad , 0.0908 rad/s , 0.99 N , 0.356 J , and 0.190 m/s , respectively.

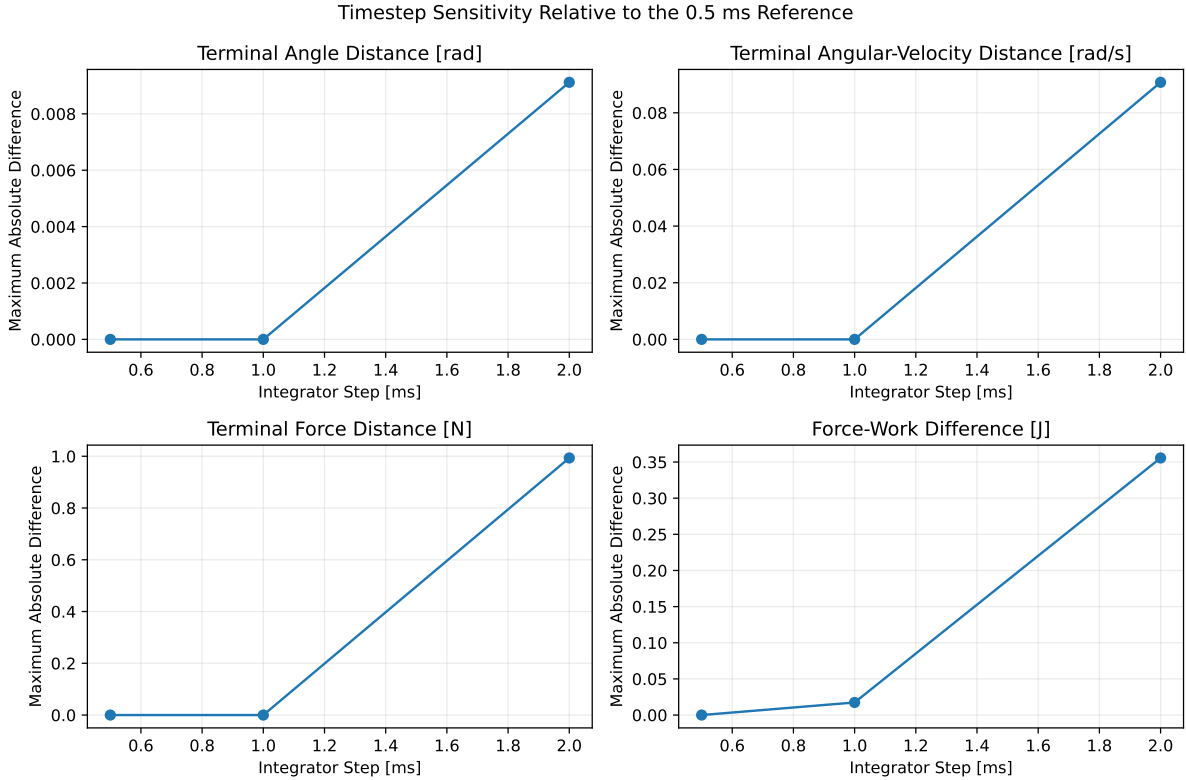


Figure 4.3: Timestep Sensitivity of Matched-State Divergence Metrics

The near-identity of the 0.5 and 1 ms state metrics supports use of the existing 1 ms article grid. Work is slightly more sensitive because it integrates a power difference and because some cases cross the idealized torque step. The 2 ms result is retained as a coarse-step warning rather than hidden by a single convergence number.

4.6 Gravity and Damping Ablations

Zero torque does not mean zero generalized forcing. Gravity and viscous damping remain inside the bias vector $b(q, \dot{q})$. To prevent the word *passive* from collapsing these mechanisms into one category, the 80 ms ensemble is repeated with projected gravity disabled and with both damping coefficients set to zero.

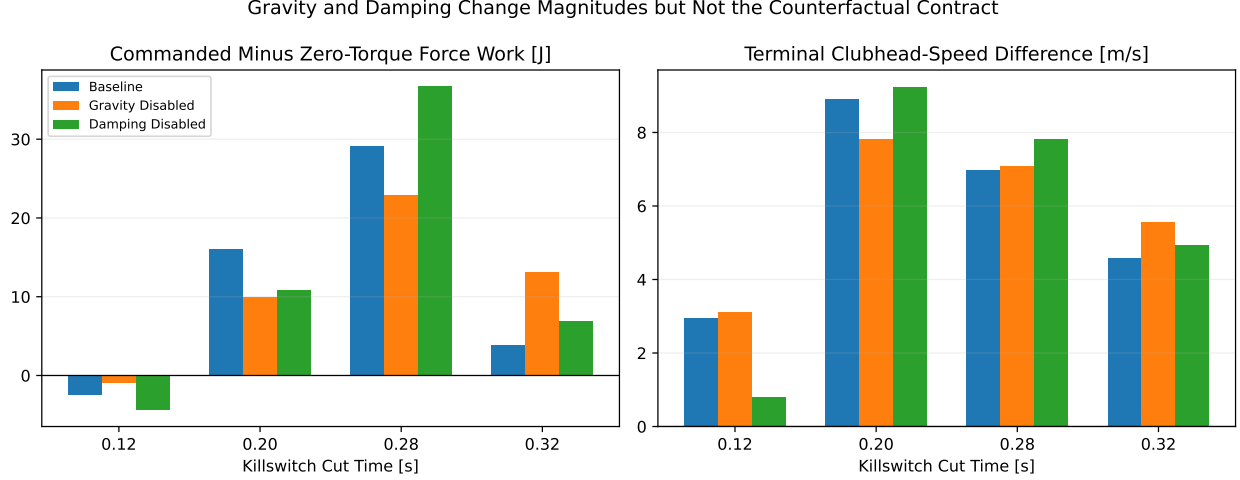


Figure 4.4: Gravity and Damping Ablations of the Killswitch Comparison

The ablations change magnitudes but do not eliminate phase dependence. At a 0.28 s cut, baseline commanded-minus-zero-torque force work is 29.17 J, versus 22.95 J without gravity and 36.73 J without damping. Terminal clubhead-speed differences remain positive in all three cases. At 0.12 s, force-work differences are negative in all three cases, while the damping-disabled speed difference is much smaller than baseline. These results show that gravity and damping materially shape the counterfactual; neither alone creates the late positive difference.

The ablations are not additive. Each altered model follows a different source trajectory before the cut, so subtracting variant numbers does not produce an orthogonal causal partition. A full mechanism partition would require matched source states with bias subterms selectively disabled inside the same dynamics evaluation. That finer separation is tracked in #8447.

4.7 WSCG BASE/ZTCF/DELTA Convention Check

The registered WSCG chart defines $\text{DELTA} = \text{BASE} - \text{counterfactual}$. Its four stored DELTA series can be reconstructed from the independently cached BASE and counterfactual series after time alignment. Maximum absolute residuals are 0.43 N for lead-hand axial, 1.48 N for lead-hand normal, 0.16 N for trail-hand axial, and 1.54 N for trail-hand normal. The residuals are small relative to the reported force ranges and arise from interpolating the dense BASE cache onto the 55-sample counterfactual grid.

This check establishes sign and subtraction convention, not model parity. The WSCG chart comes from a two-hand flexible-shaft model, while the present killswitch ensemble uses a rigid-shaft single-wrist double pendulum. Numerical agreement between their force magnitudes is neither expected nor claimed. The two-hand reproduction must compare like-for-like states, frames, hand contact points, and equivalent-wrench definitions.

4.8 Counterfactual Conclusions

The ensemble supports a stronger and more precise statement than the original pointwise analysis:

- pointwise ZTCF exactly equals the initial derivative of the matched-state zero-torque future;
- zero-torque interaction force and positive force power can persist for tens of milliseconds after a late cut;
- persistence and commanded-minus-zero-torque differences depend strongly on cut phase and horizon;
- the 1 ms integration step is converged for the reported state, force, work, and speed metrics relative to 0.5 ms; and
- gravity, damping, and the wrist-command discontinuity change quantitative results without explaining away the underlying distinction between instantaneous attribution and a realizable future.

These conclusions close the counterfactual-contract gap for the double pendulum. They do not establish the source of the two-hand negative equivalent couple. That question requires local hand forces and wrench geometry, addressed at the next fidelity level.

Chapter 5

Empirical Evidence in Golfers

5.1 The Work and Power Budget of the Swing

Nesbit’s full-body kinematic-kinetic study and the companion work-power analysis of Nesbit and Serrano remain the most complete published energy budgets of the swing (Nesbit 2005; Nesbit and Serrano 2005). Using subject-specific full-body models of four golfers of widely differing skill, they report that the trunk complex (lumbar and thoracic spine plus hips) contributes 68.7–72.2% of total swing work, the shoulder and arm joints 24.3–28%, and the legs only 3.3–3.8% (Nesbit and Serrano 2005). Two further observations bear directly on the timing question. First, power generation and clubhead speed decreased systematically with increasing handicap. Second, the phase analysis locates the *generation* of energy predominantly in the earlier downswing (torso-dominated) and the *conversion* into clubhead speed late (shoulder/arm and club interface) — generation and transfer are temporally separated, which is a precondition for the retain-early/release-late strategy to even be possible.

The ground is the ultimate source of the external impulse. Han and colleagues characterized golfer-ground interaction in 63 skilled golfers and found meaningful associations between ground-interaction force and moment parameters and maximum clubhead speed (Han et al. 2019). Recent work has pushed this to an explicitly energetic formulation: in thirty golfers (fifteen professionals, fifteen high-level amateurs), foot-ground interaction variables alone explained 35.5% of clubhead-speed variance, rising to 75.4% when trunk sequencing and an *impulse-based energy-transfer efficiency* measure (effectiveness of mechanical impulse transmission from proximal segments to the club) were added; mediation analysis attributed the dominant indirect pathway to transfer efficiency rather than to sequencing per se (Rachnavy et al. 2026). That distinction — transfer *efficiency* mattering where sequence *order* alone does not — recurs throughout the skill-level literature below.

5.2 Segmental Kinetic-Energy Sequencing

Two studies measured the quantity this document cares most about — segment *kinetic energy*, not merely angular speed — and both found the P→D structure at the energetic level. Anderson, Wright and Stefanyshyn computed segmental kinetic energies in the golf swing and documented the sequenced build-up and outward migration of energy along the chain (Anderson, Wright, and Stefanyshyn 2006). Kenny, McCloy, Wallace and Otto reproduced segmental kinetic-energy sequencing in a large-scale computer-simulated golf swing, confirming the ordered energetic handoff in a full forward-dynamic setting (Kenny et al. 2008). These studies matter here for two reasons. First, they validate the segment-energy time course as a practical observable (the same observable computed for the model in Section 9). Second, they show the energetic sequence is not an artifact of angular-velocity bookkeeping: energy itself arrives late in the club, after the proximal segments have peaked.

5.3 Kinematic-Sequence Evidence and Skill-Level Differences

The descriptive $P \rightarrow D$ findings are robust in skilled populations. Tinmark and colleagues, studying elite golfers across full and partial shots, found a significant proximal-to-distal temporal ordering of peak speeds (pelvis $\rightarrow$ torso $\rightarrow$ hands/club) with successive amplification of peak speed at every level of effort, in both men and women (Tinmark et al. 2010). Cheetham and colleagues compared amateur recreational players with touring professionals on transition- and downswing-phase kinematic sequence parameters and found professionals showed higher peak-speed magnitudes and more consistent sequencing and timing characteristics (Cheetham et al. 2008). Zheng and colleagues’ three-dimensional comparison of professional and amateur swings found systematic temporal and spatial differences, with amateurs deviating progressively more from professional norms as handicap increased (Zheng et al. 2008). Meister and colleagues provide elite rotational benchmarks (pelvis-thorax separation and its timing) against which amateurs can be compared (Meister et al. 2011), and Lindsay, Mantrop and Vandervoort review the broader catalogue of skill-level biomechanical differences (Lindsay, Mantrop, and Vandervoort 2008).

Torso-pelvis interaction variables predict performance. Myers and colleagues found upper-torso and pelvis rotation (and their separation) associated with ball speed in golfers across skill levels (Myers et al. 2008); Chu, Sell and Lephart identified a small set of biomechanical variables — among them trunk and pelvis kinematics at specific downswing events — explaining a large share of driving-performance variance (Chu, Sell, and Lephart 2010); Joyce and colleagues related three-dimensional trunk kinematics to clubhead speed across clubs (Joyce et al. 2013). On the widely coached “X-factor” specifically, Kwon and colleagues showed the computed values depend materially on methodology and that, in skilled golfers, X-factor parameters were not straightforwardly associated with clubhead velocity (Kwon et al. 2013) — another instance of a popular proxy dissolving under measurement scrutiny.

Three refinements prevent an overly tidy reading:

1. **Sequence order is common but not universal — even among the best.** Skilled golfers whose peak ordering deviates from the canonical pelvis-thorax-arm-club pattern are documented, and recent work examines how alternative sequence patterns among elite players relate to clubhead speed (Lee, Ehrlich, and Cain 2026). On-plane analyses of the “axle-chain” system in 66 skilled golfers found clubhead-speed associations concentrated in specific angular-motion characteristics rather than in gross ordering (Madrid et al. 2020).
2. **Within-player timing differences do not straightforwardly explain shot quality.** Comparing well-timed and mistimed shots *within* highly skilled players, Neal and colleagues found no significant differences in the timing lags between peak angular speeds of contiguous segments (Neal et al. 2007) — the felt sense of “timing” lives somewhere other than the inter-peak lags of a five-segment model. Movement variability is itself structured and functional in golf (Langdown, Bridge, and Li 2012; Glazier 2011), so single-trial sequence snapshots carry limited information.
3. **Method sensitivity.** As Section 2.5 noted, the identified sequence depends on component-selection methodology (Marsan et al. 2019), so cross-study comparisons of ordering statistics carry real uncertainty. Historical measurements with specialized instrumentation reached compatible conclusions about trunk-pelvis motion but with their own method dependence (McTeigue et al. 1994).

What survives these refinements: *skilled golfers generate larger proximal speeds, separate transition from release more distinctly, and transmit impulse and energy to the club more efficiently; merely exhibiting a $P \rightarrow D$ peak order is neither unique to skill nor sufficient for it* (Cheetham et al. 2008; Zheng et al. 2008; Rachnavy et al. 2026; Lee, Ehrlich, and Cain 2026). Coaching perception has partly internalized this: when surveyed systematically, coaches’ key swing parameters overlap with, but do not coincide with, the parameters the biomechanical literature supports (Smith et al. 2015).

5.4 Where the Transfer Occurs

Synthesizing Section 5.1, Section 5.2 and Section 5.3 with the mechanics of Section 2: in a skilled swing, the dominant energy *generation* occurs in the trunk-hip complex during the first two-thirds of the downswing

(Nesbit and Serrano 2005); the dominant *transfer to the club* occurs in the final phase, through the wrist interface, by a mechanism that is substantially parametric/geometric (wrist-cock release under centrifugal loading, hub-path tightening) with a modest late active wrist contribution (White 2006; Miura 2001; Sprigings and Neal 2000; Nesbit and McGinnis 2009). Less-skilled players truncate exactly this separation: early release (“casting”) moves energy into the club before proximal speed has peaked, producing the lower distal peak speeds and transfer ratios observed in amateur cohorts (Cheetham et al. 2008; Zheng et al. 2008; Milburn 1982). Chapter Section 9 reproduces this entire pattern — including the casting failure mode — inside a fully instrumented model where every energy pathway is measurable exactly.

Chapter 6

The Timing Question and the Counterfactual Framework

6.1 Two Strategies, Stated Precisely

Let $E_p(t)$ be the kinetic energy of the proximal subsystem (trunk and arms, with the club riding folded) over a downswing $t \in [t_0, t_{\text{imp}}]$, and define the transfer functional $P_w(t)$ as the inter-segment power crossing the wrist interface — joint-force power plus moment power, in the sense of Section 2.5.2. The two strategies are:

- **S1 (transfer early):** make $P_w(t)$ large as soon after transition as possible; the club begins acquiring its final energy early.
- **S2 (retain early, release late):** keep $P_w(t)$ small early (the wrist transmits centripetal force but little power) while proximal torques maximize E_p ; then allow or drive a large P_w concentrated in the final phase.

The double-pendulum, delayed-release and optimal-control literatures reviewed in Section 2 uniformly favor S2 within their model classes (Milburn 1982; Pickering and Vickers 1999; Sprigings and Neal 2000; Sprigings and MacKenzie 2002; Sharp 2009; White 2006). The mechanistic reasons compose into a coherent account:

1. **Inertia management.** With the wrist cocked, the system’s moment of inertia about the hub is small, so proximal torque buys angular velocity cheaply; early release raises the inertia while proximal torque is still accelerating the system (White 2006; T. P. Jorgensen 1999).
2. **Actuator physiology.** Torque-velocity limits mean the wrist’s small muscles add meaningful power only while wrist angular velocity is modest; early wrist torque squanders that window and can even require *negative* (restraining) torque later (Sprigings and Neal 2000).
3. **Parametric transfer needs developed centrifugal loading.** The passive outward energy flow at release, and the inward-pull gain, scale with the centrifugal field established by proximal rotation — weak early, strong late (White 2006; Miura 2001).
4. **Interaction torques do the late work for free.** Motion-dependent torques accelerate the distal segment and decelerate the proximal one automatically once release begins; the golfer’s job is to *time* this cascade, not to muscle it (Putnam 1993; Herring and Chapman 1992).

Empirically, S2-like signatures separate skill groups (Section 5.3), while the within-expert null result on inter-peak lags (Neal et al. 2007) suggests that once S2 is grossly achieved, residual performance variance lives in finer variables (release profile shape, hub-path geometry, impulse-transfer efficiency) — exactly what the counterfactual framework below is designed to expose.

6.2 What Would Falsify S2?

To keep the question scientific rather than rhetorical, we state what evidence would count against the retain-early/release-late hypothesis:

- an optimal-control solution, in a validated 3D model with physiological actuator limits, whose optimal wrist-interface power is front-loaded;
- measured expert swings in which early wrist-interface power (normalized) is *positively* associated with clubhead speed across players;
- counterfactual (ZTCF/ZVCF) analyses showing that in faster swings the control term dominates club acceleration *early* rather than late;
- **or, in the direct manipulation of Section 8: torque programs that drive the wrist early outperforming programs that retain early and drive late, under matched proximal effort.**

The last test is executed in Section 9.

6.3 ZTCF and ZVCF: Definitions

Two counterfactual accelerations are defined *pointwise* at each measured state $(q, \dot{q}, \tau)$ of a swing:

$$\ddot{q}_{\text{ZTCF}} = M(q)^{-1} (-b(q, \dot{q})), \quad (6.1)$$

$$\ddot{q}_{\text{ZVCF}} = M(q)^{-1} (\tau - b(q, \mathbf{0})) = M(q)^{-1} (\tau - g(q)), \quad (6.2)$$

where the simplification in Equation 6.2 holds because the Coriolis term (quadratic in velocity) and viscous dissipation (linear in velocity) vanish at $\dot{q} = \mathbf{0}$. The ZTCF (Equation 6.1) answers “*what would the system do right now if every actuator switched off?*” — it is the **drift** field of the affine control system, containing gravity plus all motion-dependent interaction dynamics. The ZVCF (Equation 6.2) answers “*what would the currently applied torques, together with gravity, produce from a standstill in this pose?*” — it removes every motion-dependent term but is *not* the pure control contribution, since gravity remains in Equation 6.2; the pure torque term is $M^{-1}\tau$ in the split below. Because Equation 2.1 is affine in τ , the actual acceleration superposes exactly:

$$\ddot{q} = \underbrace{M^{-1}(-b(q, \dot{q}))}_{\text{drift} = \text{ZTCF}} + \underbrace{M^{-1}\tau}_{\text{control}}, \quad (6.3)$$

an identity enforced to numerical tolerance by the accompanying tests (Section 14).

6.4 Two Operational Variants

Two distinct operationalizations are available, and the difference matters for interpretation.

Pointwise (instantaneous) counterfactuals. The engine-agnostic Python implementation ([simulation_backends/ztcf_zvcf.py](#)) evaluates Equation 6.1 and Equation 6.2 at each sample of an already-measured trajectory — explicitly *not* a forward-integrated rollout. Each sample answers a local “what if”; this is the right tool for drift/control *time profiles* along a real swing, and it is the tool used in Section 9.

Re-simulation counterfactuals. The original MATLAB/Simscape pipeline ([run_ztcf_simulation.m](#)) re-simulates the full model with a torque “killswitch” applied at time t , for a sweep of t values, tabulating the counterfactual state reached with torques off. The pipeline forms the difference table $\text{DELTA} = \text{BASE} - \text{ZTCF}$ ([process_data_tables.m](#)), attributing to active control everything the passive system would not have done. A companion ZVCF model applies the measured torques to the same pose with velocities zeroed

([SCRIPT_ZVCF_GENERATOR.m](#)); the ZVCF model is run without gravity so that, when its samples are set against the ZTCF (which retains gravity), the gravitational contribution is not double-counted. This decomposition is meaningful *pointwise*, at matched states, in the sense of Equation 6.3: the ZVCF script records one sample per zero-velocity pose, and separately integrated counterfactual rollouts do not superpose at the trajectory level in a nonlinear multibody system, because M and b evolve differently once the counterfactual state diverges. The MATLAB pipeline additionally computes, for every joint and every table (BASE, ZTCF, DELTA), the linear work $\int \mathbf{F} \cdot \mathbf{v} dt$, angular work $\int \tau \cdot \omega dt$, total work and power, and the fractional contributions ZTCF/BASE and DELTA/BASE ([calculate_work_impulse.m](#), [calculate_total_work_power.m](#)) — precisely the Robertson-Winter-style joint-interface accounting of Section 2.5.2, including a kinematic-sequence plot computed *separately for the BASE, ZTCF and DELTA components* ([SCRIPT_324_PLOT_ZTCF_KinematicSequence.m](#)).

6.5 The Counterfactual Form of the Timing Hypothesis

The ZTCF/ZVCF language makes the S2 hypothesis crisp. Along a measured downswing, project the drift and control accelerations onto the club’s speed-producing direction and integrate the associated powers. S2 predicts, for high-performing swings:

- **Early downswing:** the *control* term should dominate energy input at proximal joints, while at the wrist the control contribution to club angular acceleration should be small or negative (restraining lag against the developing centrifugal field). The club’s ZTCF profile should show the drift field already “wants” to open the wrist — control is *holding energy back*.
- **Late downswing:** the *drift* term (parametric transfer, centrifugal release) should dominate club acceleration, with a brief late window of positive wrist control power ([Springs and Neal 2000](#)). The active share of club-side work should concentrate before release; the passive share should account for most club energy gain after release ([White 2006](#)).

This pattern was initially motivated by torque-removal analyses. Section 9 provides a reproducible quantitative test in a two-segment model; conversely, had the early-drive programs won the sweep, S2 would have been falsified in exactly the terms of 3. Bounded-actuator, impact-definition, and model-parameter sensitivity checks are reported here; re-simulation parity, 3D full-body models, and measured swings remain future work.

Chapter 7

Computational Framework

The numerical work uses an open-source, backend-neutral implementation of a planar two-segment golf model. Model parameters are represented once and rendered to the analytical equations of motion and optional simulation backends. The present results use the deterministic analytical ODE backend; they do not depend on a proprietary solver or graphical interface.

7.1 Analysis Components

The workflow separates four responsibilities:

1. **Dynamics.** A validated mass matrix and bias-force provider advances the two generalized coordinates under specified shoulder and wrist torques.
2. **Counterfactual decomposition.** At each sampled state, acceleration is split into passive drift and applied-control components. These pointwise quantities are not presented as alternative forward trajectories.
3. **Energy accounting.** Segment kinetic and potential energies, actuator work, and wrist-interface force and moment power are evaluated with a Robertson–Winter balance check ([Robertson and Winter 1980](#)).
4. **Reproducibility.** Fixed-step integration, explicit parameter records, machine-readable outputs, and regression tests connect every reported number to a declared computation.

The source implementation and versioned data are linked once in the code and data availability statement (Section 12). Repository names are omitted from the scientific argument because software provenance is not evidence for the physical claims.

7.2 Validation Boundary

The implementation includes unit tests for drift-plus-control reconstruction, work–energy consistency, deterministic replay, impact detection, and headline ordering. Independent simulation backends provide a route to cross-engine validation, but the results in this document should be read as one validated implementation of one model class. Cross-engine agreement would reduce implementation risk; it would not remove structural limitations such as the fixed hub, rigid shaft, planar motion, or simplified actuators.

Chapter 8

Computational Methods

This chapter specifies the simulation experiments whose results appear in Section 9. Everything described here is implemented in three committed modules under `scripts/research/proximal_distal_energy/` (`swing_model.py`, `interaction_forces.py`, `torque_programs.py`, `run_experiments.py`, and `run_interaction_force_scripts.py` with figures from the paired figure scripts), and every output carries a provenance record (git SHA, integrator settings, parameter values). The reproduction commands are in Section 13.

8.1 Model

The simulated plant is a backend-neutral two-segment golf model (Section 7): an arm segment rotating about a fixed hub, hinged at the wrist to a club segment consisting of a distributed shaft plus a clubhead point mass at the tip, moving in an inclined swing plane with projected gravity. Baseline parameters are (`GolfModelParams.default()`; full table in Section 15): arm length 0.75 m and mass 7.5 kg; club length 1.0 m with 0.15 kg shaft and 0.20 kg clubhead; plane inclination 35°; viscous damping 0.4 and 0.25 N · m · s/rad at shoulder and wrist. Angles follow the declared convention: θ_1 is the arm angle from the downward vertical in the swing plane, θ_2 the wrist angle of the club relative to the arm, and the club’s absolute angle is $\theta_{\text{club}} = \theta_1 + \theta_2$.

Rollouts use the ODE reference backend (analytic dynamics, fixed-step RK4) with $\Delta t = 1$ ms over a 0.9 s horizon. Every rollout starts from the same top-of-backswing state, $\theta_1(0) = -2.2$ rad, $\theta_2(0) = -1.57$ rad (a 90° wrist cock), $\dot{q}(0) = 0$ — an arm carried well past horizontal with the club folded over the shoulder, comparable to the classical two-lever setups (T. P. Jorgensen 1999; Pickering and Vickers 1999).

8.2 Torque Programs

All programs share a constant shoulder torque τ_s active from $t = 0$ (two levels: 60 and 100 N · m), so the sweep isolates the wrist channel. The wrist torque follows one of three profiles, parameterized by an onset time t_{on} :

- **passive**: $\tau_w(t) = 0$ throughout (free hinge) — the White limit (White 2006);
- **drive-only**: $\tau_w(t) = 0$ for $t < t_{\text{on}}$, then $\tau_w = +15$ N · m (opening);
- **restrain-then-drive**: $\tau_w(t) = -R$ (cock-retaining) for $t < t_{\text{on}}$ with $R \in \{5, 10\}$ N · m, then $\tau_w = +15$ N · m.

The onset grid is $t_{\text{on}} \in \{0, 0.025, \dots, 0.35\}$ s (15 values). With both shoulder-torque levels this yields 92 distinct programs; **drive-only** at $t_{\text{on}} = 0$ is the S1 (“transfer early”) pole of the design and **restrain-then-drive** with late onset is the S2 pole. Torque magnitudes sit inside the ranges used by comparable two- and three-segment studies (Springs and Neal 2000; Pickering and Vickers 1999), scaled to this model’s inertias.

E1b (actuator-bound sensitivity). Because constant torques overstate what muscle can deliver at speed, a companion sweep (`e1b_bounded_torque.py`) replaces both concentric drives with a linear torque-velocity bound, $\tau(\omega) = \tau_{\text{iso}} \text{clip}(1 - \omega/\omega_{\text{max}}, 0, 1)$ (shoulder: $\tau_{\text{iso}} = 100 \text{ N} \cdot \text{m}$, $\omega_{\text{max}} = 20 \text{ rad/s}$ against arm speed; wrist: $20 \text{ N} \cdot \text{m}$, 30 rad/s against opening speed; restraint constant, since eccentric torque does not taper with shortening velocity). State-dependent torques require closed-loop integration, done with the backend’s own `mass_matrix/bias_forces` primitives under the identical RK4 settings.

8.3 Impact Definition and Validity

Impact is the first upward crossing of $\theta_{\text{club}} = 0$ (club pointing straight down in the swing plane), linearly interpolated between samples; the score of a program is the clubhead (distal tip) speed at that instant. Because the shoulder torque is constant, a program that keeps the club folded through the bottom of the swing can postpone the crossing into a second rotation while the shoulder keeps pumping energy — an artifact, not a swing. A trial is therefore **invalid** unless the arm angle at the crossing satisfies $\theta_1 \leq 2.0$ rad (a first-pass delivery); 29 of the 92 programs were excluded by this rule, all of them heavy-restraint or very-late-onset programs. This constraint is conservative *against* the S2 hypothesis: it discards only trials that would otherwise have inflated the restraint arm of the sweep.

8.4 Recorded Quantities

For every program: the full state trajectory, applied controls, impact time and clubhead speed. For four representative programs at $\tau_s = 60 \text{ N} \cdot \text{m}$ (passive; drive-only at $t_{\text{on}} = 0$; the best drive-only; the best restrain-then-drive), additionally:

Segment energies. Arm kinetic energy $\frac{1}{2} I_1^{\text{hub}} \dot{\theta}_1^2$; club kinetic energy $\frac{1}{2} m_2 \|\mathbf{v}_{c2}\|^2 + \frac{1}{2} I_2^{\text{com}} (\dot{\theta}_1 + \dot{\theta}_2)^2$ with the club centre-of-mass velocity $\mathbf{v}_{c2}$ computed analytically from the state.

Wrist-interface power accounting (E4). Following Section 2.5.2: the joint-force power $P_f = \mathbf{F} \cdot \mathbf{v}_{\text{wrist}}$, with the joint reaction force recovered from Newton’s second law for the club segment, $\mathbf{F} = m_2 \mathbf{a}_{c2} - m_2 \mathbf{g}$ (centre-of-mass acceleration by central differences of the analytic velocity); the net wrist moment power on the club $(\tau_w - b_w \dot{\theta}_2)(\dot{\theta}_1 + \dot{\theta}_2)$; the wrist actuator power $\tau_w \dot{\theta}_2$; and the residual check that the club’s total mechanical energy rate equals joint-force power plus moment power (reported as a figure in Section 9; gravity is not added separately because potential energy is included in the club’s mechanical energy).

Pointwise counterfactual split (E2). At every sample, the `drift_and_control_split` (Equation 6.3) evaluated with the applied torques, yielding drift and control components of both joint accelerations; the club’s absolute angular-acceleration split is the row sum.

Exact interaction-force audit (E5). Analytic club-centre-of-mass acceleration replaces finite differentiation and separates the wrist reaction force into proximal and distal tangential and centripetal terms plus gravity reaction (Section 3.3). Each vector is projected onto the analytic wrist velocity to obtain a signed power pathway. A late-downswing matched-state experiment then integrates the commanded and zero-torque futures for 120 ms, explicitly distinguishing a trajectory-level killswitch from pointwise ZTCF (Section 3.7). Newton balance, club moment balance, force and power reconstruction, matched initial state, and initial ZTCF equivalence are unit-tested.

Matched-state counterfactual ensemble (E6). Eight cut times, including three samples bracketing the 0.100 s wrist-command switch, are crossed with four forward horizons and three fixed timesteps. Commanded and zero-torque futures share the exact interpolated source state. Divergence is recorded in generalized position and velocity, wrist force, force power, integrated force work, and terminal clubhead speed. The 80 ms subset is repeated with gravity disabled and with damping disabled. The 0.5 ms grid is the timestep reference, and the registered WSCG chart’s stored DELTA series are independently reconstructed as BASE minus counterfactual after time alignment (Section 4).

Phase budgets. With the downswing divided at $t_{\text{imp}}/2$: club kinetic-energy gain, wrist actuator work, shoulder work, and integrated joint-force transfer, per half.

Robustness analyses (E1c and E1d). E1c re-scores the same 92 trajectories using fixed arm positions, peak horizontal clubhead velocity, peak first-pass speed, a matched arm angle, and five validity bounds. E1d repeats the four-strategy comparison after changing one model input at a time: arm length ($\pm 10\%$), arm mass ($\pm 15\%$), club length ($\pm 10\%$), clubhead mass ($\pm 20\%$), swing-plane inclination (25° and 45° versus 35°), and both damping coefficients ($0.5\times$ and $1.5\times$). For each E1d case, late-drive and restrain-then-drive onset are re-optimized on the original onset grid. These ranges test local structural sensitivity; they are not population percentiles or confidence intervals.

8.5 Design Rationale and Scope

This deliberately simple model is a planar 2-DOF chain with a fixed hub and rigid shaft. That choice is the point, not a compromise: the classical timing results were established in exactly this model class (Section 2.3), so the analysis first reproduces the class’s behavior under explicit instrumentation — where every energy pathway is exactly measurable and the counterfactual contract is tested — before scaling to models where the same questions are harder to instrument. What the 2-DOF results can and cannot establish is taken up explicitly in Section 10; the extension pathway (re-simulation parity, higher-fidelity models, and motion capture) is specified in Section 11.

Chapter 9

Results

All numbers in this chapter are computed by the committed experiment suite (Section 8) from the recorded outputs in [data/](#) (`e1_sweep.json`, `representative_traces.npz`, `results_summary.json`); figures are generated by `make_figures.py`. Speeds are quoted to 0.1 m/s and times to 1 ms.

9.1 E1 — Timing of the Distal Handoff

Figure 9.1 shows clubhead speed at impact as a function of wrist-drive onset time for both shoulder-torque levels. Table 9.1 lists the representative programs.

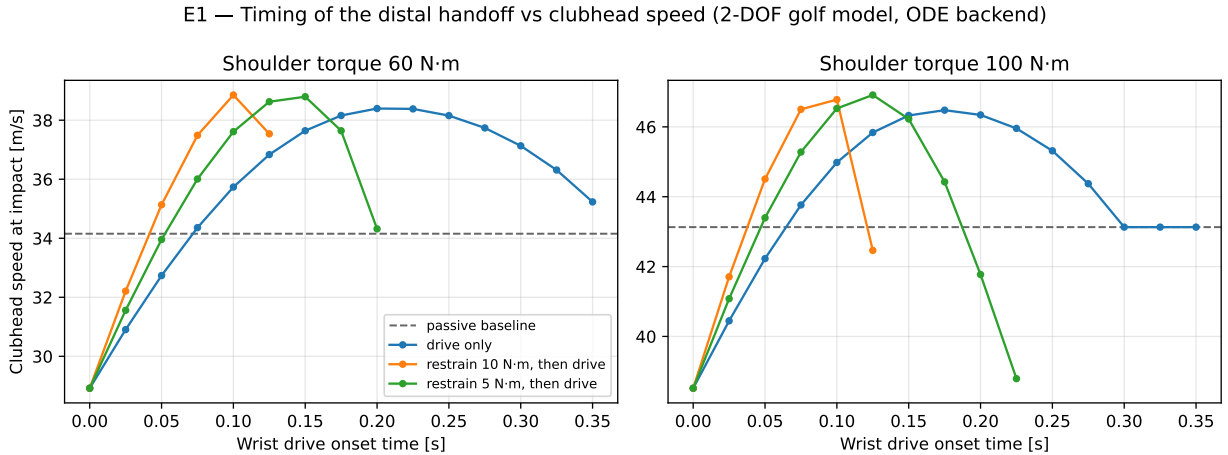


Figure 9.1: E1 sweep: clubhead speed at impact versus wrist-drive onset time, for drive-only and restrain-then-drive profiles at two shoulder-torque levels. Dashed line: fully passive wrist baseline.

Table 9.1: Representative programs at $\tau_s = 60 \text{ N} \cdot \text{m}$.

Program ($\tau_s = 60$ $\text{N} \cdot \text{m}$)	t_{on} [s]	t_{imp} [s]	Clubhead speed [m/s]	vs passive
Passive wrist	—	0.370	34.2	—
Early drive (S1 pole)	0.000	0.327	28.9	−15.3%
Best drive-only	0.200	0.352	38.4	+12.4%

Program ($\tau_s = 60$ N · m)	t_{on} [s]	t_{imp} [s]	Clubhead speed [m/s]	vs passive
Best restrain-then-drive ($R = 10$)	0.100	0.349	38.9	+13.8%

Three findings, consistent across both torque levels:

1. **Driving the wrist from the top is the worst valid strategy.** At $\tau_s = 60$ N · m the early-drive program reaches 28.9 m/s versus 34.2 m/s for a completely passive wrist — active early distal effort *destroys* 15% of the speed a free hinge would have delivered. The early-drive program is the worst valid trial in the entire 92-program sweep. At $\tau_s = 100$ N · m the pattern repeats (early drive is the minimum of its profile curve; passive baseline 43.1 m/s).
2. **The same wrist torque, applied late, is worth +12%.** Clubhead speed rises monotonically with onset time until an interior optimum ($t_{\text{on}} = 0.200$ s at 60 N · m, 0.175 s at 100 N · m — roughly the middle of the downswing, with drive active through the final ~40%), reaching 38.4 and 46.5 m/s respectively.
3. **Retain-early-then-drive is best of all.** Programs that apply a *negative* (cock-retaining) wrist torque early and the identical positive drive late beat the best drive-only programs at both torque levels: 38.9 m/s ($R = 10$ N · m until 0.100 s) at $\tau_s = 60$, and 46.9 m/s ($R = 5$ N · m until 0.125 s) at $\tau_s = 100$. The margin over the best drive-only program is modest (+0.5 m/s at both levels), consistent with most of the retention benefit being achievable passively; but the ordering is strict, and heavy or over-long restraint fails the first-pass validity rule
(2) rather than winning — the benefit is a *timed* retention, not maximal retention.

9.2 E1b — Sensitivity to Actuator Torque-Velocity Bounds

A standing objection to constant-torque sweeps is that real muscle torque falls with shortening velocity, shrinking the late-drive window (Sprigings and Neal 2000). E1b re-runs the sweep with a linear torque-velocity bound on both concentric drives (shoulder: 100 N · m isometric, zero at 20 rad/s of arm speed; wrist: 20 N · m isometric, zero at 30 rad/s of opening speed; restraint left constant, as eccentric torque does not taper with shortening velocity), integrated closed-loop with the same RK4 settings (Figure 9.2). The ordering survives intact: early drive 28.8 m/s < passive 30.8 < best late drive 34.6 (onset 0.175 s) < best restrain-then-drive 34.8 ($R = 10$ N · m until 0.100 s). Effect sizes compress — late drive is worth +12.2% over passive and the restraint margin narrows to +0.2 m/s — precisely the compression bounded-actuator theory predicts, while the optimal onsets move slightly *earlier* (0.175–0.100 s versus 0.200–0.100 s), since a velocity-limited wrist needs a slightly longer window to deliver its work. The S2 ordering is therefore not an artifact of unbounded actuators.

9.3 Kinematic Sequence

Figure 9.3 shows arm and club angular-speed traces. All four programs produce the qualitative P→D pattern — arm speed peaks, then club speed peaks higher — as expected from interaction dynamics alone (Section 2.2). The programs differ exactly where the skill literature says they should (Section 5.3): the early-drive swing opens the club immediately, its arm peak is depressed, and its club peak is the lowest of the four; the retain-then-drive swing holds the club folded longest, achieves the highest arm speed before release, and converts it into the highest club peak. Impact in the early-drive swing occurs with the arm still 0.36 rad *short* of vertical — the clubhead has overtaken the hands before the arm arrives, the model’s version of “casting” / premature release — whereas the two best programs deliver impact with the arm 0.7–0.8 rad *past* vertical (hands ahead of the clubhead, forward shaft lean), matching the geometry of skilled impact.

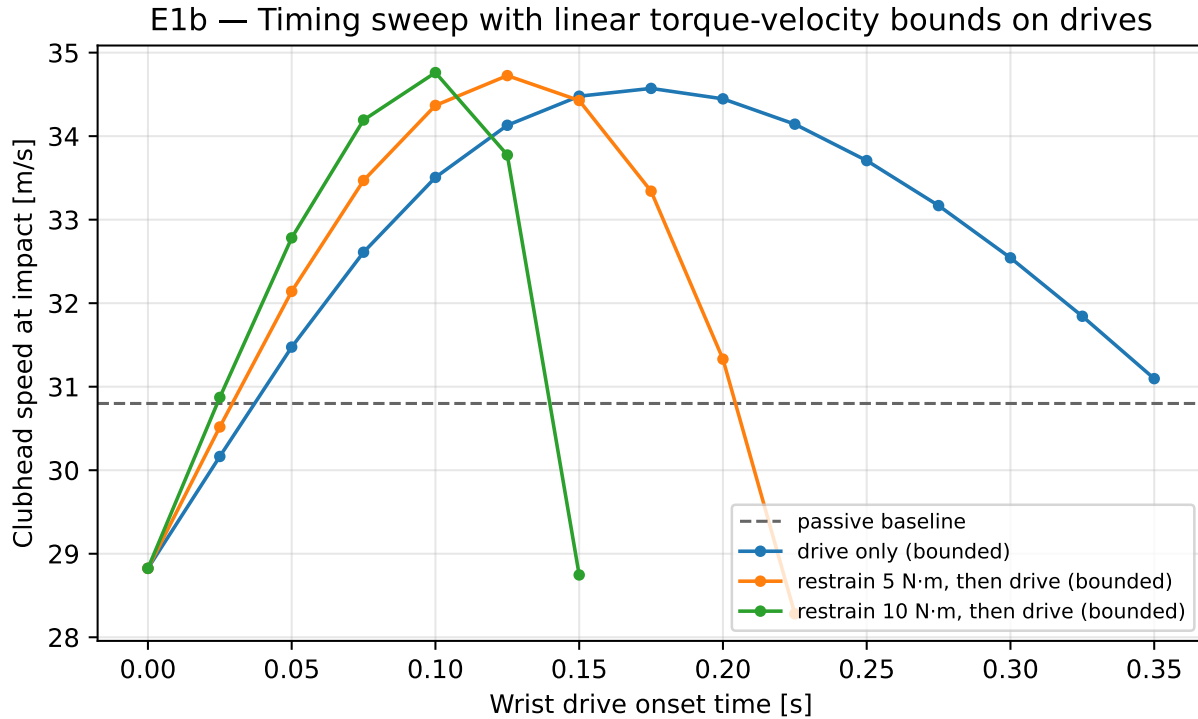


Figure 9.2: E1b: the timing sweep repeated with linear torque-velocity bounds on the shoulder and wrist drives.

9.4 Segment Energies

Figure 9.5 shows arm and club kinetic-energy time courses, and Figure 9.6 the early-half/late-half budget. The energetic story is unambiguous:

- **Early half.** The winning programs put the *least* energy into the club early: 5.0 J (best restrain) and 7.3 J (best drive-only) versus 22.9 J for the early-drive program (passive: 7.8 J). Meanwhile they bank more energy in the arm: early-half shoulder work is 54.7 J (best restrain) and 51.9 J (best drive) versus 33.9 J (early drive) — the early-driven club back-reacts on the arm and *steals its runway*.
- **Late half.** The ordering inverts: late club kinetic-energy gain is 180.5 J (best restrain) and 174.2 J (best drive) versus 84.4 J (early drive) and 136.4 J (passive).
- **Active retention is measurable.** The best restrain program’s early-half wrist actuator work is *negative* (−1.9 J) and its early-half joint-force transfer is likewise slightly negative (−1.0 J): the wrist interface is momentarily a net energy *withdrawer* from the club while the proximal system loads — the literal, quantitative meaning of “retaining energy proximally”.

9.5 E4 — Wrist-Interface Power Accounting

Figure 9.7 (left) shows the Robertson-Winter interface powers for the best restrain-then-drive swing. The joint-force power — energy carried into the club *through the joint force*, the pathway that requires no wrist torque at all — is small and briefly negative early, then surges in the final ~0.1 s to dominate the club’s energy supply (late-half integral 131.7 J versus 25.7 J of late wrist actuator work). The wrist moment channel contributes a brief late pulse, exactly the “latter stages” window identified by Sprigings and Neal (Sprigings and Neal 2000). The right panel verifies the accounting: the club’s mechanical-energy rate tracks the sum of joint-force and moment power across the swing (residual at the level of the finite-difference approximation),

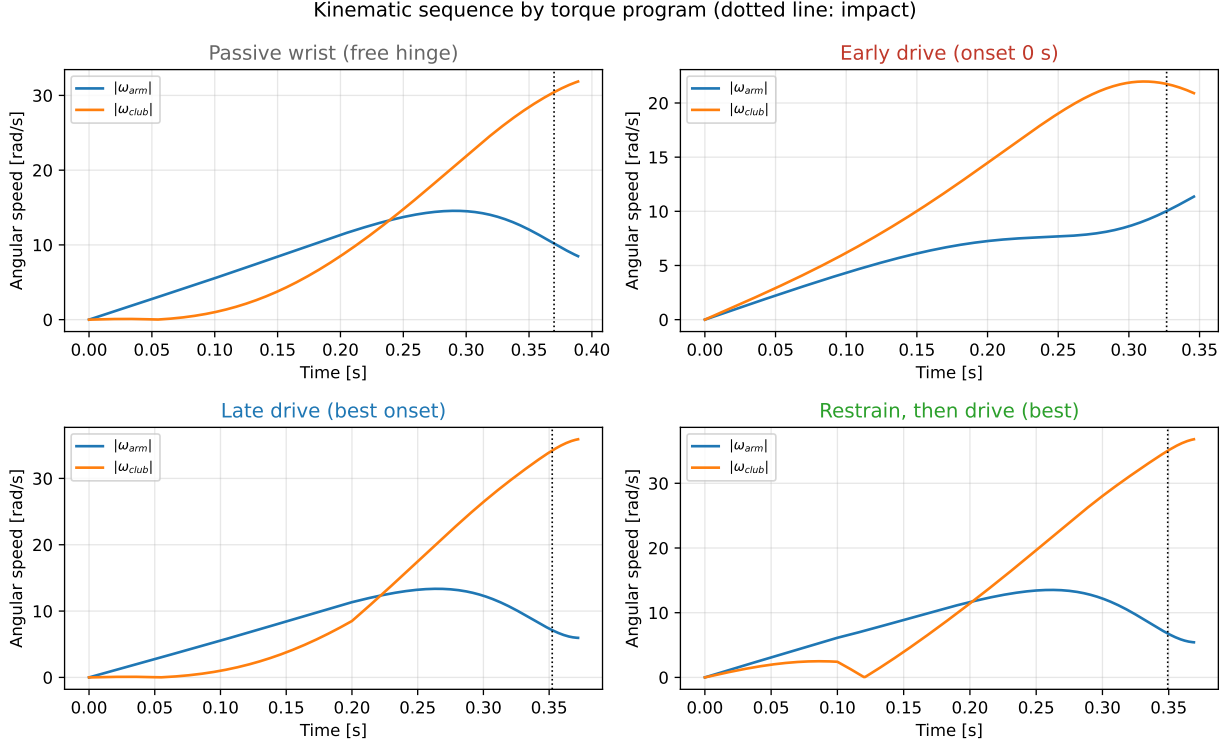


Figure 9.3: Kinematic sequence (arm and club angular speeds) for the four representative programs at $\tau_s = 60$ N · m. Dotted vertical line: impact.

confirming the implementation of the segmental balance of Section 2.5.2.

9.6 E2 — Counterfactual Drift/Control Split

Figure 9.8 shows the pointwise drift/control decomposition (Equation 6.3) of the club’s absolute angular acceleration for the S1 pole (early drive) and the S2 winner (restrain-then-drive). In the early-drive swing, the control term supplies a large share of club acceleration from the first instant — the S1 signature — and the drift term never develops the late surge seen elsewhere. In the restrain-then-drive swing the picture reverses: early, the control term *opposes* the drift term at the club (restraint holding back a drift field that already wants to release); late, the drift term — the ZTCF, i.e. what the accumulated motion would do with every actuator switched off — dominates the club’s acceleration entirely, with the control term a small additive assist. This is, instant by instant, the counterfactual signature that Section 6.5 predicted for high-performing swings: **early control retains; late drift delivers**.

9.7 E1c and E1d — Robustness to Scoring and Model Parameters

Re-scoring all trajectories under five alternative families of impact criteria preserved the qualitative strategy ordering. This rules out the specific concern that the result was created by the club-vertical impact definition, but it does not test unmodeled impact physics.

The one-at-a-time parameter study also preserved the ordering in all 13 cases. Across those cases, clubhead-speed ranges were 26.4–31.3 m/s for early drive, 32.9–35.6 m/s for the passive wrist, 36.5–40.6 m/s for the best late drive, and 36.8–41.3 m/s for the best restrain-then-drive program. The best drive onset shifted from 0.175 to 0.225 s, while the best restrained onset shifted from 0.100 to 0.150 s.

Restrain, then drive (best): downswing progression in the swing plane

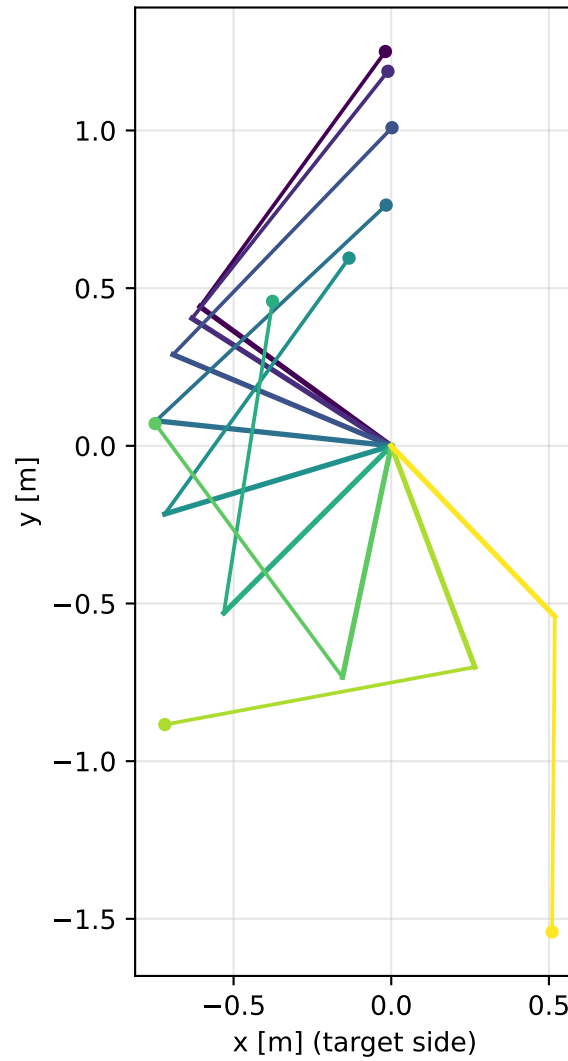


Figure 9.4: Stick-figure montage of the best restrain-then-drive swing in the swing plane (downswing progression at equal time steps).

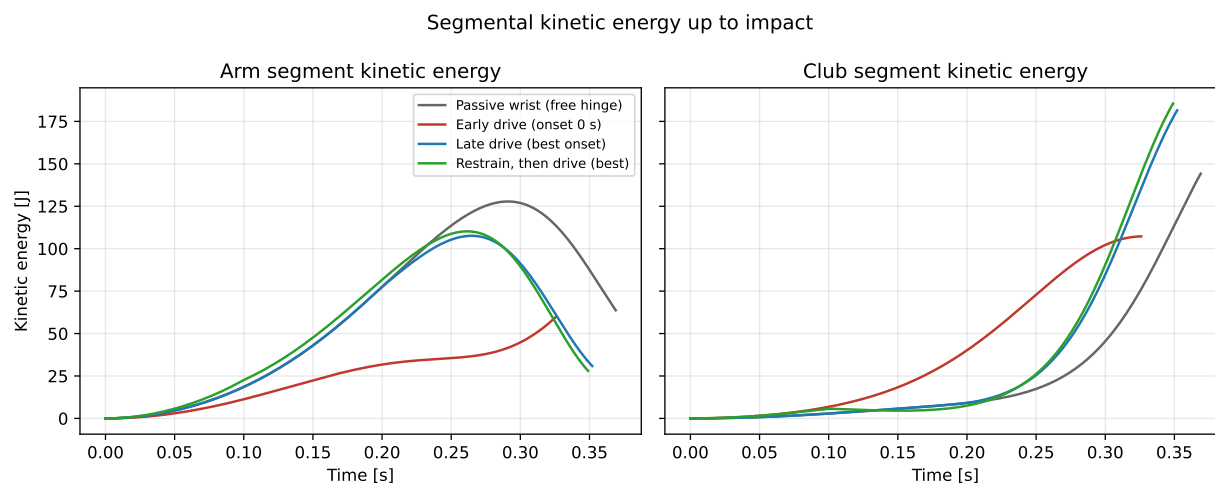


Figure 9.5: Segmental kinetic energies up to impact for the four representative programs.

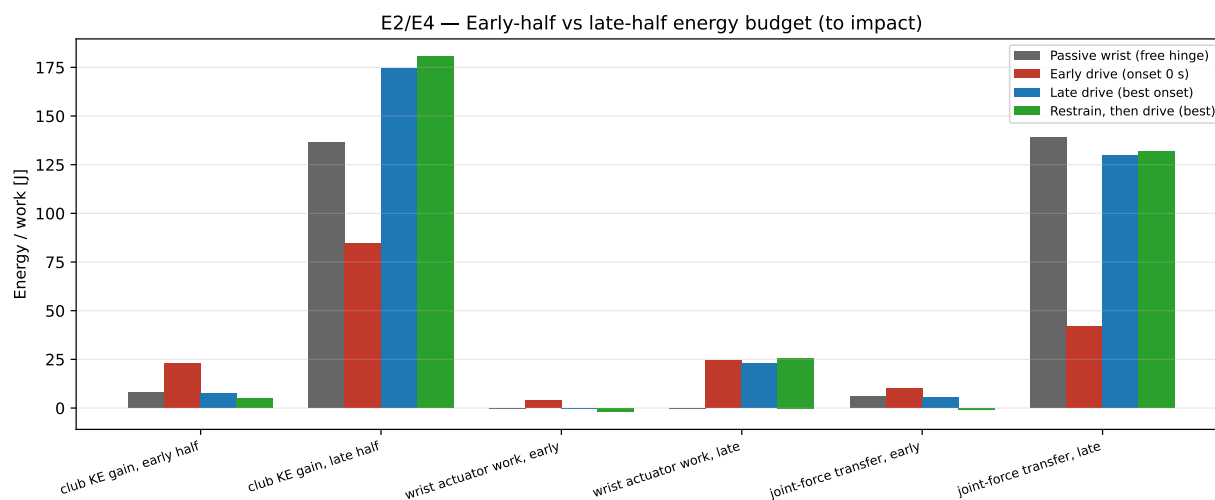


Figure 9.6: Early-half versus late-half energy budget (club kinetic-energy gain, wrist actuator work, integrated joint-force transfer) for the representative programs.

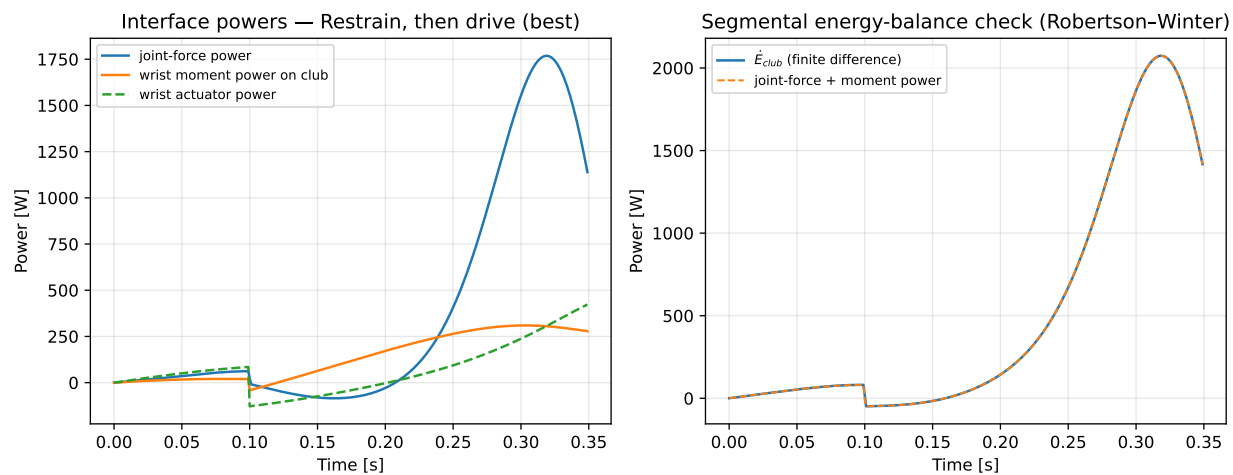


Figure 9.7: Left: wrist-interface powers for the best restrain-then-drive swing. Right: segmental energy-balance check — club mechanical-energy rate versus the sum of interface powers.

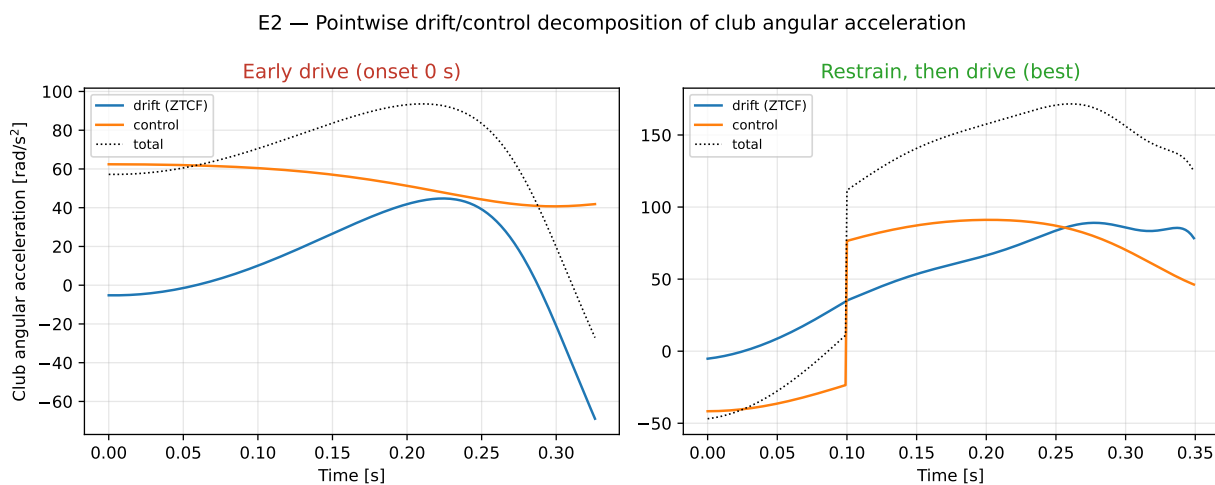


Figure 9.8: Pointwise drift (ZTCF) versus control decomposition of club absolute angular acceleration, for the early-drive (left) and best restrain-then-drive (right) programs.

The stable ordering supports a mechanism-level interpretation within the local parameter neighborhood. The shifting optimum argues against treating one onset time as a general prescription. Because parameters were varied individually, the study does not characterize interaction effects or a population distribution.

9.8 Summary of Findings

Within this model class, under matched proximal effort and a consistently defined first-pass impact: (i) early distal drive is lower than the passive condition (-15% in the baseline case); (ii) identical distal effort applied late is worth $+12\%$; (iii) timed active retention followed by late drive is best at both effort levels, with its early-phase wrist work measurably negative; (iv) the late club energy supply is dominated by the passive joint-force/parametric pathway, not the wrist actuator; (v) the counterfactual split shows control acting early as restraint and drift delivering late — the S2 pattern of Section 6.5 in full; and (vi) the entire ordering survives linear torque-velocity bounds, alternative scoring definitions, and the declared one-at-a-time parameter perturbations, although effect sizes and optimal onset times change (Section 9.2, Section 9.7).

Chapter 10

Discussion

10.1 Agreement With the Literature

Every headline result of Section 9 has an antecedent in the literature of Section 2, and the quantitative agreement is close enough to be meaningful rather than merely directional. The harm of early release reproduces the delayed-release results of Pickering and Vickers and of Sprigings and MacKenzie (Pickering and Vickers 1999; Sprigings and MacKenzie 2002) and Milburn’s original observation that early uncocking truncates arm acceleration (Milburn 1982). The +12% value of late wrist drive brackets Sprigings and Neal’s 4% gain from a physiologically bounded late wrist torque (Sprigings and Neal 2000) — our unbounded constant 15 N · m torque and longer active window make a larger gain expected. The dominance of the passive late energy pathway (joint-force transfer of 131.7 J versus 25.7 J of late wrist work in the best swing) is White’s parametric transfer made explicit in Robertson-Winter accounting (White 2006; Robertson and Winter 1980). The casting geometry of the early-drive swing — clubhead overtaking the hands before the arm reaches vertical — is the model’s rendering of the amateur release failure documented across skill-level studies (Cheetham et al. 2008; Zheng et al. 2008), and the impact geometry of the winning programs (hands ahead, arm 0.7–0.8 rad past vertical) matches the skilled pattern. Finally, the drift/control time signature — control as early restraint, drift as late delivery — is the counterfactual pattern proposed in Section 6.5, now exhibited by this model.

10.2 Why Retention Wins: A Mechanistic Reading

The energy budgets localize the causal structure. Early wrist drive fails through *two* simultaneous channels: it raises the system’s moment of inertia about the hub while the shoulder is still accelerating (so the same shoulder torque buys less arm speed — the early-drive program banks 21 J less early shoulder work than the retain program despite identical τ_s), and it spends the wrist’s work budget at a time when the joint-force/parametric channel cannot yet amplify it (White 2006). Timed restraint wins through the mirror mechanisms: it holds the compact configuration slightly longer than a free hinge would (the drift field already wants to open the wrist — the restraint is fighting centrifugal release, not gravity), banks that margin as arm speed, and then releases into a fully developed centrifugal field where the passive channel converts arm energy to club energy at the maximum rate. The negative early wrist work (−1.9 J) is small in absolute terms but diagnostic: retention is an *energetic withdrawal*, not merely an absence of drive — precisely the “restraining torque” role that appears in Jorgensen’s fitted expert swing and in Sprigings-Neal-type actuator solutions (T. P. Jorgensen 1999; Sprigings and Neal 2000; Herring and Chapman 1992).

Two honest qualifications temper the S2 reading. First, the margin of active restraint over the best *passive-then-drive* program is small (+0.5 m/s at both effort levels): in this model most of the retention benefit is available from simply *not driving* the wrist, with active holding adding a final increment. Second, retention is only valuable *timed*: heavier or longer restraint programs failed the first-pass validity rule rather than

winning — over-retention postpones delivery past the useful window. “Retain early, release late” is therefore not “retain maximally”; it is *retain until the drift field can pay, then stop retaining*.

10.3 What the 2-DOF Model Cannot Establish

The results chapter demonstrates the S2 pattern in the model class where the classical timing literature lives; it does not yet demonstrate it for a golfer. The specific gaps, each with a tracked remediation:

- **Fixed hub, rigid shaft, planar chain.** The hub-path and inward-pull mechanisms (Miura 2001; Nesbit and McGinnis 2009; Nesbit and McGinnis 2014), shaft dynamics (MacKenzie and Sprigings 2009; Betzler et al. 2012; Osis and Stefanyshyn 2012), and long-axis rotations (Marshall and Elliott 2000) are all outside this model. Closed-loop, flexible-shaft, and full-body models are appropriate staged extensions.
- **Simplified actuator models.** E1b shows the S2 ordering survives a linear torque-velocity bound on the drives with compressed margins (Section 9.2), but activation-rate limits, history dependence, and eccentric-side detail remain unmodeled; full optimal-control treatments with physiological actuators are the appropriate refinement (Sprigings and Neal 2000; Sharp 2009; McNally and McPhee 2018).
- **Local, one-at-a-time sensitivity.** Alternative impact definitions and single-parameter perturbations preserved the ordering (Section 9.7), but coupled parameter variation, measurement uncertainty, and population variability were not modeled.
- **Pointwise counterfactuals only.** The Python pathway evaluates instantaneous ZTCF profiles; the MATLAB re-simulation variant (killswitch rollouts, DELTA tables) has not yet been reproduced in Python, so the two operationalizations have never been formally compared on matched trajectories.
- **No human data yet.** The decisive observational test — wrist interface power time courses across skill levels from motion capture
(3) — awaits an appropriately governed motion-capture study.

10.4 Implications, Cautiously Stated

For modeling practice, the analysis shows that the timing question is tractable with counterfactual decomposition, and that energy-level accounting (not peak ordering) is the observable that separates strategies — supporting the methodological argument of Section 2.5 and the transfer-efficiency findings in recent human data (Rachnavy et al. 2026). For coaching interpretation, the results align with instruction that discourages “hitting from the top” and encourages retained lag with late release (Cochran and Stobbs 1968; T. P. Jorgensen 1999) — but a 2-DOF torque-program result is evidence about mechanism, not a prescription for any individual player, whose constraints (mobility, strength, consistency, injury history (Cole and Grimshaw 2016; Hume, Keogh, and Reid 2005)) sit outside this model. Any coaching translation should also respect that within-player timing variability in experts is real and partly functional (Neal et al. 2007; Langdown, Bridge, and Li 2012; Glazier 2011).

Chapter 11

Conclusions and Research Priorities

11.1 Conclusions

The literature does not support equating proximal-to-distal peak order with a requirement to transfer mechanical energy outward as early as possible. Interaction dynamics, optimal-control studies, and segmental energy measurements instead motivate a timing hypothesis: retain a compact proximal configuration while proximal speed develops, then accelerate the distal handoff later in the downswing.

The present simulations provide model-derived support for that mechanism. Under matched shoulder torque in a planar two-link model, early wrist drive reduced clubhead speed relative to a passive wrist, whereas optimized late drive increased it. A small early restraining torque followed by late drive was best among the tested programs. Energy and counterfactual accounting located the modeled mechanism in early arm-energy retention and late joint-force transfer rather than in early active wrist work.

This conclusion is conditional. The result survived torque-velocity bounds, alternative scoring rules, and 13 one-at-a-time parameter cases, but the optimal onset shifted across cases. The study therefore supports a robust ordering within a local model neighborhood, not a universal timing value or an individual coaching prescription. A fixed hub, rigid shaft, planar chain, simplified actuators, and absence of human data remain material limitations.

11.2 Research Priorities

The next work should reduce the largest inferential risks in dependency order:

1. **Counterfactual-method parity.** Implement torque-killswitch re-simulation in Python, compare it with the pointwise decomposition on matched trajectories, and cross-check the existing MATLAB pathway. Pointwise attribution and forward divergence answer different questions; agreement should not be assumed.
2. **Coupled sensitivity and constrained control.** Replace one-at-a-time perturbations with a declared multivariable design, add activation-rate and history-dependent actuator limits, and compare preselected torque programs with a constrained optimal-control solution.
3. **Model-fidelity ladder.** Repeat the primary comparisons in a torso–arm–club chain, then add mobile-hub and flexible-shaft energy terms before moving to closed-loop and full-body models. Each rung should carry conservation checks and cross-engine tolerance gates.
4. **Human-data protocol.** Predefine inclusion criteria, outcomes, uncertainty treatment, and inverse-dynamics sensitivity before analyzing appropriately licensed or consented motion and force data. Sequence order alone must not be treated as evidence of energy transfer.
5. **Open-resource release.** Maintain deterministic commands, machine-readable results, data dictionaries, checksums, citation metadata, and explicit license boundaries. Archive validated releases with a

persistent identifier.

The implementation roadmap and acceptance criteria are tracked in [the open research epic](#).

Code and Data Availability

Analysis code, tests, figures, machine-readable outputs, and parameter records are available in the [open-source repository](#). The repository name identifies provenance only; none of the scientific claims depend on adoption of the broader software project. The principal summary is recorded in [data/results_summary.json](#), and the parameter-sensitivity results are recorded in [data/eid_parameter_sensitivity.json](#).

Chapter 12

Software and Data Availability

The open-source implementation separates the analytical model, torque programs, counterfactual evaluation, energy accounting, robustness analyses, and figure generation. The scientific entry points are collected under [scripts/research/proximal_distal_energy/](#), with shared model parameters and dynamics under [simulation_backends/](#). This is a provenance statement, not a capability claim about the surrounding software.

The current evidence boundary includes these known limitations:

- A multi-phase matched-state torque-killswitch ensemble is complete for one declared torque program; other torque programs, model classes, and uncertainty distributions remain necessary before making general persistence claims.
- The model has two planar links and a fixed hub; torso, hub-path, long-axis rotation, and closed-loop arm effects are absent.
- The shaft is rigid, so strain-energy storage and recoil are absent.
- Actuator bounds are simplified; muscle activation and history effects are not represented.
- Parameter sensitivity is one-at-a-time; coupled uncertainty and population distributions are not estimated.
- No human motion or force data are analyzed; the result is mechanistic and model-derived, not a skill-group effect.

Chapter 13

Reproducibility Guide

From the repository root (Python 3.11+ with the project dependencies):

```
# Primary sweep, counterfactuals, and interface-power analysis
python3 -m scripts.research.proximal_distal_energy.run_experiments

# Registered WSCG source extraction and interaction-force analysis
python3 -m scripts.research.proximal_distal_energy.extract_wscg_charts
python3 -m scripts.research.proximal_distal_energy.run_interaction_force_study
python3 -m scripts.research.proximal_distal_energy.run_counterfactual_ensemble

# Robustness analyses
python3 -m scripts.research.proximal_distal_energy.e1b_bounded_torque
python3 -m scripts.research.proximal_distal_energy.e1c_impact_sensitivity
python3 -m scripts.research.proximal_distal_energy.e1d_parameter_sensitivity

# Figures and document
python3 -m scripts.research.proximal_distal_energy.make_figures
python3 -m scripts.research.proximal_distal_energy.make_interaction_force_figures
python3 -m scripts.research.proximal_distal_energy.make_counterfactual_figures
cd docs/research/proximal_distal_energy_transfer
quarto render proximal_distal_energy_transfer.qmd --to pdf
```

Outputs land in **data/** (JSON and NPZ with commit, integrator, and parameter provenance) and **figures/** (PDF and SVG). The analyses use fixed-step RK4 and contain no stochastic elements. Re-runs should reproduce the recorded values up to floating-point environment differences.

Chapter 14

Validation Evidence

Tests exercise analytical double-pendulum ground truth, optional cross-backend agreement, drift-plus-control reconstruction, work–energy consistency, deterministic replay, impact edge cases, headline values, and the parameter-case contract. The wrist-interface balance also checks that the club mechanical-energy rate matches the summed interface powers to the finite-difference tolerance. These checks reduce numerical and implementation error; they do not validate the structural model assumptions.

Chapter 15

Model Parameters and Program Definitions

Table 15.1: Baseline model parameters and experiment constants.

Parameter	Value	Notes
Arm length l_1	0.75 m	baseline
Arm mass m_1	7.5 kg	both arms lumped
Arm COM ratio	0.45	$l_{c1} = 0.3375$ m
Arm inertia about COM	$0.3516 \text{ kg} \cdot \text{m}^2$	about hub: $1.2059 \text{ kg} \cdot \text{m}^2$
Club length l_2	1.0 m	grip to clubhead
Shaft mass	0.15 kg	COM ratio 0.43
Clubhead mass	0.20 kg	point mass at tip
Club composite mass m_2	0.35 kg	$l_{c2} = 0.7557$ m
Club inertia about COM	$0.0403 \text{ kg} \cdot \text{m}^2$	about wrist: $0.2402 \text{ kg} \cdot \text{m}^2$
Plane inclination	35°	projected gravity 8.033 m/s^2
Damping (shoulder, wrist)	0.4, 0.25 $\text{N} \cdot \text{m} \cdot \text{s/rad}$	viscous
Integrator	RK4, $\Delta t = 1$ ms	analytical ODE backend
Initial state	$\theta_1 = -2.2, \theta_2 = -1.57$ rad, $\dot{q} = 0$	top of backswing
Shoulder torque τ_s	60 / 100 $\text{N} \cdot \text{m}$	constant from $t = 0$
Wrist drive	+15 $\text{N} \cdot \text{m}$ from t_{on}	onset grid 0–0.35 s, step 0.025
Wrist restraint R	5 / 10 $\text{N} \cdot \text{m}$ before t_{on}	restrain-then-drive only
Impact	first $\theta_1 + \theta_2 = 0$ crossing	valid iff $\theta_1 \leq 2.0$ rad

Recorded outputs:

- `data/e1_sweep.json` — one row per torque program plus provenance.
- `data/representative_traces.npz` — representative state, control, counterfactual, energy, speed, and interface-power traces.
- `data/results_summary.json` — representative impacts and phase budgets.
- `data/e1b_bounded_sweep.json` — torque-velocity-bounded sweep.
- `data/e1c_sensitivity.json` — alternative impact-definition results.
- `data/e1d_parameter_sensitivity.json` — 13 one-at-a-time parameter cases.
- `data/interaction_force_mechanisms.npz` — exact force, power, geometry, and matched-state kill-switch arrays.
- `data/interaction_force_summary.json` — headline force, work, and divergence metrics with counterfactual contracts.

- `data/wscg_2024_hand_force_series.csv` — hash-verified values from the source presentation’s embedded chart cache.
- `data/wscg_2024_source_provenance.json` — source hashes, series inventory, and interpretation boundary.
- `data/counterfactual_ensemble.json` — 96 baseline cut/horizon/timestep rows, gravity and damping ablations, convergence metrics, and WSCG DELTA checks.
- `data/counterfactual_selected_traces.npz` — commanded and zero-torque state, force, and power traces for three representative cuts.

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